Faculty of Computer Science and Information Technology

TYRE TREAD WEAR INDICATOR ANALYSIS USING DIGITAL IMAGE PROCESSING

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Bachelor of Computer Science with Honors

(Computational Science)

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This project is submitted in partial fulfillment of the
requirements for the degree of
Bachelor of Computer Science with Honors
(Computational Science)

Faculty of Computer Science and Information Technology
UNIVERSITI MALAYSIA SARAWAK
2018

ANALISIS INDIKATOR TAPAK TAYAR MENGGUNAKAN PEMPROSESAN IMEJ DIGITAL

NOOR SAHERA BINTI ROSLI

Projek ini merupakan salah satu keperluan untuk

Ijazah Sarjana Muda Sains Komputer Dengan Kepujian

(Sains Komputan)

Fakulti Sains Komputer dan Teknologi Maklumat

UNIVERSITI MALAYSIA SARAWAK

2018

DECLARATION

I hereby declare that the thesis of Tyre Tread Wear Indicator Analysis Using Digital Image Processing is based on my original work except for quotations and citations which have been duly acknowledging. I also declare that no portion of the work referred to in this report has been submitted in support of an application for another degree at Universiti Malaysia Sarawak (UNIMAS) or qualification of this or any other university or institution of higher learning.

_____

Noor Sahera Binti Rosli (47994) 17 May 2018
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ABSTRACT

Car's tyre is the most important equipment in the vehicle. To increase the friction and safety of the vehicle, variant treads are in the surface of the tyre. However, the friction and safety are decreased while the tyre is worn down to the tyre indicator and when the tread has become shallow. Worn tyres will result in slippage, longer braking distance to stop on the slick road and can lead to an accident. Therefore, governments, including Malaysia, have set standard measurement for depth tread of the tyre must be greater than 1.6 mm. Consequently, this study proposed to determine the algorithm in digital image processing which can be used to measure the depth tread of the tyre with the use of vision inspection. This study will use three different algorithms in research experiment and analyze the result of measure of the depth tread of the tyre.

ABSTRAK

Tayar kereta adalah peralatan terpenting dalam kenderaan. Untuk meningkatkan geseran dan keselamatan kenderaan, pelbagai tapak tayar dicipta di atas permukaan tayar. Walau bagaimanapun, geseran dan keselamatan akan berkurangan, apabila tayar yang dipakai haus sehingga ke penunjuk tayar dan apabila tapaknya menjadi cetek. Tayar yang dipakai akan mengakibatkan kemerosotan, jarak pengereman yang lebih lama untuk berhenti di jalan yang licin dan boleh menyebabkan kemalangan. Oleh itu, kerajaan, termasuk Malaysia, telah memutuskan pengukuran standard bagi ketebalan tayar mestilah lebih besar daripada 1.6 mm. Oleh itu, kajian ini dicadangkan untuk menentukan algoritma dalam pemprosesan imej digital yang boleh digunakan untuk mengukur jejak kedalaman tayar. Kajian ini akan menggunakan tiga algoritma yang berlainan dalam percubaan penyelidikan dan akhirnya, satu dari tiga algoritma akan dipilih sebagai algoritma untuk mengukur jejak kedalaman tayar.

CHAPTER 1: INTRODUCTION

1.1 INTRODUCTION

Nowadays, accidents have been identified as a leading factor of death in Malaysia. According to Road Safety Department Malaysia (RSD), there were 6 700 deaths reported from road fatalities in the year 2009 and 15% of them were caused by faulty vehicles tyres such as worn tyre (Mohamed Idris, 2010). Every vehicle has an adequate tyre tread depth that can help to improve the contact of the tyre with the road while maintaining the grip on the road in the variety of conditions such as rain through the groove of the tyre tread. If the tyres are worn down to the tyre wear indicator, it may take so much time to stop on the slick road and can lead to an accident. Tyres and safety experts believe that the minimum legal tyre tread for cars in Malaysia is 1.6 mm (Tan, 2014).

In order to decrease the numbers of vehicles accident caused by tyres, every car must follow the legal measurement of the tyre tread with an approximation of depth tread of the tyre must greater than 1.6 mm. A proper inspection should be carried out to make sure the depth tread of tyre follows the standard safety measurement. Even though conventional technique exists for instance coins test, depth gauge and indicator mark this technique is not efficient as the computerized inspection. Thus, despite using this conventional technique to inspect the tyres, computerized inspection by using digital image processing is more efficient and effective to inspect the depth tread of the tyre compared to the conventional technique.

In computer science and information technology field, image processing is a form of signal processing. The input of image processing can be digital images or video while the output of the image processing will be the characteristics related to the input. Most of the image processing is referring to digital image processing (Anbarjafari, 2017). The digital image processing techniques use the computer algorithm to create, process, communicate and display the related digital images (George & Rhoades, 2015). Advantageously, the digital image processing can improve the visual appearance of images compared to human visual, manipulation the digital images by using different algorithm, analysis the images of tyres in more sharpness and better visual appearance, improve the conventional technique to inspect

the tyres with computerized inspection and can perform analysis on images even though the tyres has been covered by external factor such as mud without need to clean the tyres.

The main purpose of this project is to research on algorithm that able to measure the depth tread of the tyre by using digital image processing. This research will attempt to make the first step by collecting ten different data or images of tread wear of tyre which consists of 5 images of used and new tyres in image acquisition stage. In image processing phase, the image will be converted into the format that is readable by MATLAB software. Few conversion format processes will be performed on the image by converting the image from RGB (Red, Green, Blue) format into greyscale. Followed by image processing, effective techniques such as running programming algorithms is used to process the digital image. This study will use three different algorithms in research experiment and finally, one out of three algorithms will be chosen as the algorithm to measure the depth tread of the tyre. Algorithms such as Histogram, Edge Detection and Hausdorff Distance algorithm are the three chosen algorithms that will be used to measure the depth tread of the tyre. The decision to propose algorithm to measure the depth tread of the tyre will be decided after the experimentation phase.

The proposed algorithm can help in determining the depth tread of the tyre. With the use of digital image processing to measure depth tread of the tyre, this computerized inspection will able to reduce the number of the accident by following the legal tyre tread or standard safety measurement. It was hoped that the result gathered through this project will able to solve the problem and the proposed algorithm can be used in the implementation of mobile development or web-based in the future which can help the user in determining the depth tread of the tyre in fast and accurate computerize inspection.

1.2 PROBLEM STATEMENT

The tread depth of tyre can be easily overlooked when life is getting busy or when we more focused on other things. However, heavily tread worn, or proper tread depth is the important key to help maximize user safety and performance on the road. The conventional technique to measure the depth tread of tyre still relies on the visual inspection for instance by using coins test, depth gauge, and indicator mark. However, this type of inspection is not very noticeable,

inefficient and unsafe to the user which has higher chance or probability of the misjudgment and human error in reading in getting the accurate measurement of tread wear of the tyre.

This research addresses this problem and it will assist by proposing the best algorithm in digital image processing to measure the depth tread of tyre by using computerized inspection and give the accurate output in the percentage after performing the analysis of image processing towards depth tread of tyre.

In the era of information technology, computerized inspection by using digital image processing can perform safety measurement more accurate and efficient. Therefore, the digital image processing technique is a good approach which can be used in measuring depth tread of the tyre.

1.3 SCOPE

Initially, the scope of this project is to measure the depth tread of the tyre by comparing the output of three different algorithms in the digital image processing. Due to the limitation of time, the scope of this project is necessary to be limited. Therefore, the scope of this project is limited to car tyres in Malaysia only. This project will not undergo investigation on other features of tyres such as the type of grooves or to test tyre on other condition such as wet grip test condition. The data of image tread of the tyre will not depend on the same type of grove, the model of tyre and cars. At the same time, MATLAB will be the only tools used for digital image processing. Finally, this project will be focusing only on digital image processing to measure the depth tread of the tyre.

1.4 OBJECTIVE

The main objective of this research is to research on algorithm to measure the depth tread of the tyre by comparing three set of algorithms in digital image processing which can analyse the depth tread of the tyre better by using digital image processing in the efficient and effective way. This project allows giving an accurate percentage of depth tread of the tyre compared to conventional technique which has been using nowadays.

Other objectives of this project include:

- To differentiate the depth tread between the used and the new tyres.
- To propose an algorithm to measure the depth tread of the tyres by comparing 3 set of algorithms in digital image processing.
- To find out the best technique or algorithm to analyses the depth tread of the tyre by using digital image processing.
- To give an accurate measurement of depth tread of tyre measurement by using digital image processing.

1.5 BRIEF METHODOLOGY

Research methodology is a method used to successfully achieve all the research objective and solved the research problem systematically. This research-based project will use qualitative research method. Qualitative research is a scientific research that was used to seeks answers or solve the problem statements with a set of procedure.

Therefore, the methodology has been divided into few steps:

Figure 1.1: Methodology step in this project

PHASE 1: RESEARCH QUESTION AND BACKGROUND

During the initiation phase, the question of the research question is constructed by discussing with supervisor to identify the everything needed to conduct the experiment in this research-based project.

PHASE 2: DATA COLLECTION

All the data for the research should be collected before research experimentation phase. In this phase, the data is collected by capturing ten different images of tyre tread from different cars and groove by using DSLR camera (Digital single-lens reflex). There will be ten images of tyre tread consist of five images of the new and used tyre. During this process, the car registration plate will be recorded for analysis purposes. The image will be converted into JPEG (Joint Photographic Expert Group) for image analysis and the image resolution will set to 1100 x 1100 pixel. Few conversions will be performed by converting the RGB (Red, Green, Blue) format into greyscale images.

PHASE 3: RESEARCH EXPERIMENTATION

In research experimentation phase, the data image between the new tyre and the used tire will be computed by using three different algorithms by using MATLAB. The algorithms that will be used is Histogram, Edge detection and Hausdorff Distance algorithm. Different algorithm will compute a different result. Therefore, the result of the three algorithms will be compared with each other in order to find the finest algorithm which will able to give the desired output in measuring the depth tread of the tyre.

PHASE 4: RESULT AND CONCLUSION

After research experimentation is conducted by comparing three algorithms, the result of each algorithm will be recorded. To determine which algorithm to choose, the results of the three algorithms in research experimentation will be listed in the table and be compared which each other. The result that indicated the highest percentage will be chosen as the algorithm to determine the depth tread of the tyre in digital image processing.

PHASE 5: DOCUMENTATION

In the last phase of methodology, documentation is required to present the information and give details after the research has been carried out. The report will also recommend the proposed algorithm that can be used to determine the depth tread of the tyre.

1.6 SIGNIFICANCE OF PROJECT

The primary target that will achieve throughout this project are:

- Impact on the study:
    - Provide an accurate measurement in term of percentage when the depth tread of the tyre is inspected by using digital image processing which will increase the level of accuracy of inspection of the car tyre.
    - Present an algorithm for digital image processing to measure the depth tread of the tyre.
    - Present a new way to inspect the tyre by using computer vision of digital image processing in more effective and efficient way.
    - Documentation of tyre tread wear indicator using digital image processing can be done after the investigation is settled.

- Impact of society:
    - The statistic number of the accident which caused by the tyre is worn down will decrease with the presence of the new method to inspect the tyre by using digital image processing.
    - Provide safety standard for vehicles in Malaysia.

1.7 PROJECT SCHEDULE

Table 1.1: Project Schedule for Final Year Project 1 & Final Year Project 2

TASK NAME	DURATION	START	FINISH
Final Year Project 1	**100 days**	**Mon 11/9/17**	**Fri 26/1/18**
FYP Briefing	**25 days**	**Mon 11/9/17**	**Fri 13/10/17**
Finding Supervisor	3 days	Mon 11/9/17	Wed 13/9/17
Finding FYP topic	2 days	Thu 14/9/17	Fri 15/9/17
Discussion with Supervisor	1 day	Mon 18/9/17	Mon 18/9/17
Brief Project Description Work Process	5 days	Tue 19/9/17	Sat 23/9/17
Submission of Brief Project Description	1 day	Sun 24/9/17	Sun 24/9/17
Approval and Comments by Reviewers	1 day	Sat 30/9/17	Sat 30/9/17
Discussion of Full Research Proposal with SV	1 day	Tue 3/10/17	Tue 3/10/17
Research Proposal Work Process	7 days	Wed 4/10/17	Thu 12/10/17
Submission of Full Proposal to FYP system	1 day	Fri 13/10/17	Fri 13/10/17
Chapter 1	**13 days**	**Sat 14/10/17**	**Tue 31/10/17**
Chapter 1 Work Process	12 days	Sat 14/10/17	Mon 30/10/17
Submission of Chapter 1	1 day	Tue 31/10/17	Tue 31/10/17
Chapter 2	**18 days**	**Wed 1/11/17**	**Fri 24/11/17**
Chapter 2 Work Process	17 days	Wed 1/11/17	Thu 23/11/17
Submission of Chapter 2	1 day	Fri 24/11/17	Fri 24/11/17
Chapter 3	**16 days**	**Sat 25/11/17**	**Fri 15/12/17**
Chapter 3 Work Process	18 days	Tue 21/11/17	Thu 14/12/17
Submission of Chapter 3	1 day	Fri 15/12/17	Fri 15/12/17
Submission of Full FYP 1 Report to FYP System	1 day	Tue 26/12/17	Tue 26/12/17
Final Year Project Symposium	2 days	Wed 3/1/18	Thu 4/1/18
Submission of Final report	1 day	Fri 26/1/18	Fri 26/1/18
Final Year Project 2	**90 days**	**Mon 12/2/18**	**Fri 15/6/18**
Revise Proposal and Gantt Chart	5 days	Mon 26/2/18	Fri 2/3/18
Chapter 4	**30 days**	**Sat 3/3/18**	**Fri 13/4/18**
Chapter 4 Work Process	30 days	Sat 3/3/18	Thu 12/4/18
Submission of Chapter 4	1 day	Fri 13/4/18	Fri 13/4/18
Chapter 5	**11 days**	**Sat 14/4/18**	**Mon 30/4/18**
Chapter 5 Work Process	12 days	Sat 14/4/18	Sun 29/4/18
Submission of Chapter 5	1 day	Mon 30/4/18	Mon 30/4/18
FYP 2 Report Draft & Paper Draft for Symposium	13 days	Tue 1/5/18	Thu 17/5/18
Final Year Project Symposium	3 days	Wed 23/5/18	Fri 25/5/18
FYP 2 Final Report & Paper After Amendment	6 days	Mon 28/5/18	Mon 4/6/18
Submission of Final Report, source code & Paper for Assessment	8 days	Tue 5/6/18	Thu 14/6/18

1.8 EXPECTED OUTCOME

At the end of this project, thesis and a paper about the investigation of the tyre tread wear indicator using digital image processing will be produced. The new technique to measure depth tread of the tyres by using computerized inspection or system will be produced as well as the result of the research experimentation that will show a best technique or algorithm to use in measuring the depth tread of the tyre. Besides, from this result, the technique or algorithm can be proved effective. Overall, this project will broaden our understanding towards the algorithms in the digital image processing by using MATLAB. This project will able present the best algorithms to be used in measuring the depth tread of the tyre while in the computational field, this project will lay a knowledge base for digital image processing.

CHAPTER 2: LITERATURE REVIEW

2.1 INTRODUCTION

The writing of literature review in Chapter 2 can be elucidated as the process of reading, analyzing, evaluating and summarizing about the currently existing system or technique of a particular topic which is related to the research study. In order to achieve the objective and requirement of this project, related studies of numerous existing system or technique have been carried out. The literature review will hinge on numerous source for instance articles, books, journals and website to gather an overview technique to measure the depth tread of the tyre as well to identify the strength and weakness of the existing system or technique. The findings will be assembled and compared with the proposed technique to measure the depth tread of the tyre by using digital image processing.

They are many researchers have proposed and developed numerous technique to measure the depth tread of the tyre which available in the market nowadays starting from the conventional tyre depth measurement to the current technology measurement of the tyre tread depth. However, there is some requirement that needs to be highlighted in order to fulfill user needs.

In this chapter, the various technique for measure the depth tread of the tyre is described in the following section. This chapter begins with the background of the research study, review of existing techniques used to measure the depth tread of the tyre, followed by the comparison of all reviewed technique.

2.2 BACKGROUND

As stated earlier in Chapter 1, accidents have been identified as a leading factor of deaths in Malaysia and 15% of them was caused by faulty vehicles tyre. The increment of the number of the accident is such a worrisome as it increased day by day and involved all ages of Malaysian's nations. There was numerous strategy that has been carried out to reduce the number of accidents, for instance, the road safety campaigns.

However, despite conducting numerous road safety campaigns to reduce the number of accident, tyre tread measurement with the helped of image processing can be an effective technique to reduce the number of accidents caused by the worn tyre. The tyre tread measurement technique can accurately measure the percentage of depth tread of the tyre and alert the user to retain a tread depth of tyre in the range of standard safety measure which at least 75% of the non-interrupted band or no less than 1.6mm of tread depth.

Even though the conventional techniques to measure the depth tread of tyre, for instance, the depth gauge is existed and available in the market, this technique is not very convenient for computerizing inspection in term of output. In the conventional technique, the user must use the tool to inspect the tyre manually by themselves. The output given by depth gauge might differ from the computerize inspection. Thus, tyre tread measurement by using image processing is a good computerize inspection that allows the camera to take the images of tyres, process the images by using the algorithm in image processing and give an accurate percentage of tyre tread depth automatically to the users.

Computerized inspection by using digital image processing can perform safety measure more accurate and efficient while providing a very safe vehicle's condition on tyre compared to the conventional measurement. This solution is more protective and introduces safer tyre conditions that can help in reducing the numbers of accidents happened every day caused by worn tyres.

2.3 REVIEW OF THE EXISTING TECHNIQUES

There are several system and technique that was used to measure the depth tread of tyre starting from the conventional tyre tread depth measurement to the current technology measurement for tyre tread depth, for instance, tyre tread depth gauges and measurement of tyre tread depth by using image triangulation, light-spot analysis, and texture analysis. Several existing techniques were used to measure the depth tread of the tyre have been reviewed for several factors and reported in this subsection.

2.3.1 CONVENTIONAL TECHNIQUE OF TYRES TREAD DEPTH

Before the implementation of technology technique of tyre tread depth measurement, the conventional technique, for instance, tyre tread depth gauge is used to measure the depth tread of the tyre. According to the Tire Tech (n.d), tread depth gauge can easily be used by pushing the tread depth against any hard-flat surface and push the measuring scale into the gauge as far as possible. To measure the depth tread of the tyre, the probe must be placed into the center of a circumferential of tyre groove and push down the gauge's base and by removing the gauge by holding the barrel the tread depth reading can be measured.

Figure 2.1: Conventional Tyres Tread Depth Gauge

The conventional measurement technique allows the user to inspect the tyre manually by using the depth gauge tool. The scale of depth gauge indicates the output of the depth tread of tyre measurement. However, as the conventional technique depends on visual inspection by reading the scale of the depth gauge. This type of inspection is not very noticeable, inefficient and unsafe to the user which has higher chance or probability of the misjudgment and human error in reading in getting the accurate measurement of the depth tread of the tyre as mentioned in Chapter 1 compared to computerize inspection by using image processing.

2.3.2 MEASUREMENT OF TYRES TREAD DEPTH WITH IMAGE TRIANGULATION

- **Introduction**

Huang, Chen, and Wang (2016) proposed the measurement of depth tread of the tyre by using image triangulation. This study measures the depth tread of the tyre by using non-contact measurement prototype system which applies machine vision that use image triangulation to measure the tread depth which helps to reduce the manpower expense and improve the efficiency of tread depth measurement. In this technique, the algorithm used is Semi-Global Matching (SGM) and LabVIEW stereo vision as the tool. LabVIEW stereo vision is used to measure the percentage of depth tread of the tyre and use the image processing of edge detection to determine the correct depth in the tread's region of interested (ROI). This proposed system also allows the user to acquire information regarding the depth tread of the tyre through the Internet by connecting to a database with the Internet of Things (IoT). The user can monitor the depth tread of the tyre via website or Android phone as the system store the record in MySQL database which is programmed in PHP.

- **System Design**

In this research study, two identical cameras using image triangulation were place in parallel position on the camera base while four light sources were placed on each side of four sides camera base as shown in Figure 2.2.

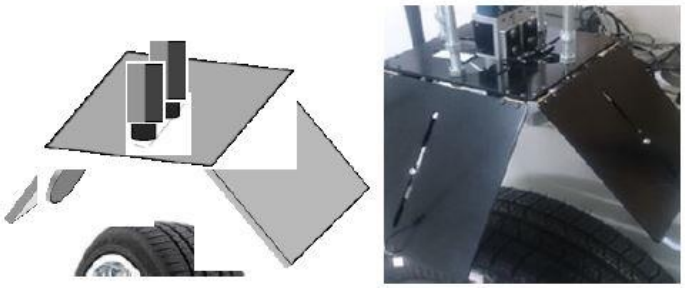


Figure 2.2: Real Stereo Vision Environment

As mentioned by Huang, Chen, and Wang (2016), the image of depth tread of the tyres can be obtained after the images from the two identical cameras were calculated by using the stereo vision function of LabVIEW. The percentage of depth tread of tyres will be calculated

by analyzing the relative distance between the surface and groove by using traditional image processing, for instance, edge detection to determine the depth tread of the region of interested (ROI) and Fuzzy C-Mean (FCM). Throughout the journal, Epipolar plane as shown in Figure 2.3 was used to obtain the distance between the tyre treads and camera lens, as a result, the depth image of tyre tread was constructed.

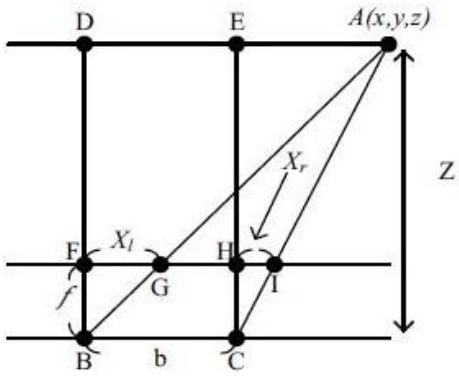


Figure 2.3: Epipolar plane

- **Tyres Tread Algorithm**

Semi-Global Matching (SGM) algorithm in LabVIEW is used to measure the distance between the camera and the tyre tread that follows the Epipolar plane concept. Followed by transferring the field view of cameras to a depth image where the pixels value indicates the corresponding points between the tire tread and the lens. In addition, the depth tread of the tyre is calculated by analyzing the relative distance between the surface and groove of the tyre inside the ROI as shown in Figure 2.4.

Figure 2.4: Region of interested (ROI) in the tyre image

ROI will analyze and obtain the tread depth by subtracting the depth at the edge of the left tread surface and depth at the bottom of the groove. As the pixel value of depth at the edge of the left tread surface might be slightly different by using SGM function, an outlier pixel in

depth images was filtered by using a median filter while the depth images were enhanced by using an image adjustment function in MATLAB. The traditional Fuzzy C-Mean (FCM) method was used to combine the pixel at the same value. To calculate the percentage of depth tread of the image, the distance obtained between the tyre tread and lens by using Epipolar plane concept and depth between the surface and the groove of main tyres found in depth images by using SGM is used. The depth tread of the tyre was calculated according to the depth at the left tread surface and depth at the bottom of the groove.

- **The Experimental Result**

The experiment results in this research were obtained by adjusting the brightness level and wind size. The experiment was conducted with the brightness level of 4000 lux and 7500 lux. The wind size of LabVIEW stereo vision of SGM algorithm is adjusting to numerous different of wind size in order to determine the influence of the wind size on the measured depth tread of the tyre. The experiment used a new tyre (Table I) and two old worn tyres (Table II and III). Each surface of the tyres was divided into eight equal parts and an image of the tyre tread was captured at the center of each part. The experiment result indicates that better result was obtained by analyzing the tyre tread images with the brightness level of 7500 lux under image triangulation and different disparity of the wind size.

TABLE I
LUX7500 NEW TIRE DEPTH

Wind Size	5	7	9	11	13	15	17	19
New Tread (a)	8	8	8	8	8	8	8	8
New Tread (b)	8	8	8	8	8	8	8	8
New Tread (c)	9	8	8	8	8	8	8	8
New Tread (d)	3	9	9	9	9	9	9	9
New Tread (e)	9	9	9	8	8	8	8	8
New Tread (f)	9	9	9	9	8	8	8	8
New Tread (g)	9	8	8	8	8	8	8	8
New Tread (h)	8	8	8	8	8	8	8	8

The actual tire tread depth was 8 mm.

TABLE II
LUX7500 OLD-1 TIRE DEPTH

Wind Size	5	7	9	11	13	15	17	19
Old-1 Tread (a)	4	4	4	4	4	4	4	4
Old-1 Tread (b)	4	4	4	4	4	4	4	4
Old-1 Tread (c)	4	4	4	4	4	4	4	4
Old-1 Tread (d)	4	4	4	4	4	4	4	4
Old-1 Tread (e)	4	4	4	4	4	4	4	4
Old-1 Tread (f)	4	4	4	4	4	4	4	4
Old-1 Tread (g)	4	4	4	4	4	4	4	4
Old-1 Tread (h)	4	4	4	4	4	4	4	4

The actual tire tread depth is 4 mm.

TABLE III
LUX7500 OLD-2 TIRE DEPTH

Wind Size	5	7	9	11	13	15	17	19
Old-2 Tread (a)	5	5	5	5	5	5	5	5
Old-2 Tread (b)	5	5	5	5	5	5	5	5
Old-2 Tread (c)	4	4	4	4	4	4	4	4
Old-2 Tread (d)	5	5	5	5	5	5	5	5
Old-2 Tread (e)	5	5	5	5	5	5	5	5
Old-2 Tread (f)	5	5	5	5	5	5	5	5
Old-2 Tread (g)	5	5	5	5	5	5	5	5
Old-2 Tread (h)	5	5	5	5	5	5	5	5

The actual tire tread depth is 5 mm.

Figure 2.5: Experiment Result with the brightness level of 7500 lux

2.3.3 TEXTURE ANALYSIS TO INSPECT THE TREAD WORN STATUS ON THE TYRES

- **Introduction**

Huang and Syu (2014) proposed the Texture Based Region of Interested (TBRI) algorithm to evaluate the region of interested (ROI) which restrained on specific texture that is suited to evaluate the depth tread of the tyre by using Discriminative Robust Local Ternary Pattern (DRLTP). This proposed technique used computer vision technology or computerize inspection to inspect the status of the tread of the tyre. Based on the experimental result, DRLTP can evaluate the status of tread worn which in the ROI that extracted by the proposed algorithm TBRI.

- **Extract Tread Texture**

The status of tread worn of the tyre is indicated between the distance of the new tread of the tyre and the worn tread of the tyre. The extraction of specific tread texture from numerous types of tread texture is an important procedure, therefore, TBRI algorithm is proposed to evaluate the ROI that restrained the specific textures as shown in Figure 2.6. The ROI was divided into the horizontal border of ROI (Coarse ROI) and the vertical border of ROI (Fine ROI).

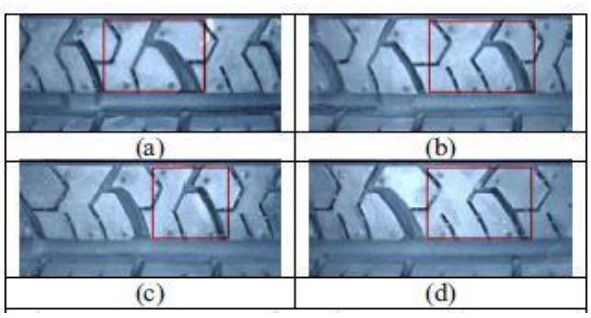


Figure 2.6: Four types of tyre tread texture

- **Texture Evaluated by DRLTP**

Discriminative Robust Local Ternary Pattern (DRLTP) is used to evaluate the status of tread worn in the ROI that extracted by the proposed algorithm TBRI. Advantageously, DRLTP can retain contrast information between the new and worn tread and robust to the image variation caused by the inversion of intensity. The inversion of intensity is divided into upper

and lower abbreviated as Upper Local Binary Pattern (ULBP) and Lower Local Binary Pattern (LLBP) respectively as shown in Figure 2.7.

$$ULBP_C = \sum_{b=0}^{B-1} f(p_b - p_c)2^b \qquad LLBP_C = \sum_{b=0}^{B-1} g(p_b - p_c)2^b$$

Figure 2.7: Equations of ULBP and LLBP

P_c is noted as the grey value of the center pixel. P_b is noted as the grey value at the border of the mask. The pre-processing of DRLP involved are a segment and edge filter. The segment processing which segments the grey image into smooth and edge cluster. The smooth cluster *T*, is the difference between p_c and the maximum pixel grey value. Edge filter converts the grey images to the edge image. Consequently, the grey image is converted to ULBP and LLBP image where each point value composed as U = $\{u_i\}$, and L=$\{l_j\}$. Based on ULBP and LLBP, four histograms which are UR, LR, UD, LD is expressed to form DRLTP as shown in Figure 2.8.

$$DRLTP(j) = \begin{Bmatrix} UR(u_{j-1}) & , & 1 \le j \le m \\ LR(l_{j-m-1}) & , & (m+1) \le j \le m+n \\ UD(u_{j-m-n-1}) & , & m+n+1 \le j \le 2m+n \\ LD(l_{j-2m-n-1}) & , & 2m+n+1 \le j \le 2m+2n \end{Bmatrix}$$

Figure 2.8: Four histograms to from DRLTP

- **Experimental result**

In this study, three tyres of the same type are used. This experiment uses 78 images of the new tyre and 200 images of a worn tyre. After that, TBRI algorithm was applied to extract the major tread as shown in Figure 2.9. However, in this experiment one of ROI is failed to extract out of 78 images of the new tyre and 33 of ROI are failed to extract out of 200 images of a worn tyre. ROI may have failed to extract single, two, loss or incomplete major tread.

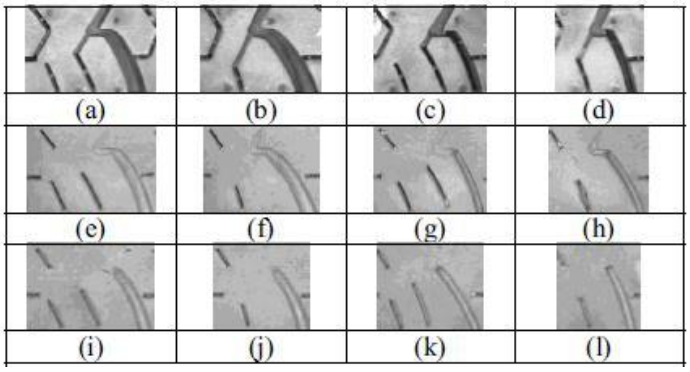


Figure 2.9: Extracted ROI of new tread (a)-(d) and worn tyre (e)-(l)

In this experiment, 68 ROI of the new tread of the tyre is used to generate the template histogram where the mean value of ROI_n is defined in figure 1. The status of depth tread of the tyre is evaluated by E_i, the rate difference between the template histogram GH(j) and the ith wore tread histogram $DRLTP_i$ as defined in Figure 2.10.

$$GH(j) = \frac{\sum_{k=1}^{p} DRLTP_k(j)}{P} \qquad E(i) = \frac{\sum_{j=1}^{2m+2n} |GH(j) - DRLTP_i(j)|}{\sum_{j=1}^{2m+2n} GH(j)}$$

Figure 2.10: Equations of GH(j) and E(i)

Based on Figure 2.11 image (e)-(l) ROI correspond to Fig 12 is the 22nd, 83th, 53th, 9th, 128th, 102th, 135th, 166th ROI respectively. In the graph, E22=0.367, E128=0.403 that correspond to Figure 2.9 (e)-(i). As the tyre worn heavily, the value of E_i will be larger.

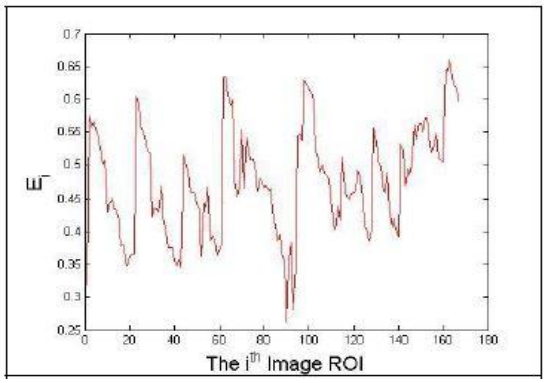


Figure 2.11: The status of the worn tread of tyre

2.3.4 MEASURING TREAD DEPTH WITH LIGHT-SPOT ANALYSIS

- **Introduction**

Wang and Huang (2017) proposed to measure the depth tread of tyre accurately by using non-contact measuring depth method which is based on the laser light-spot analysis. Throughout this journal, in order to get an experimental result of depth tread of tyre accurately, the tyre will undergo an analysis by using laser light-spot and effective analysis of these laser light-spot is used to measure the depth tread of tyre accurately.

- **Laser Light-Spot Difference**

Due to the angled exposure and flatness of the exposed surface, the laser light-spot will appear differently under vertical exposure and uneven surface. In this study, the laser light-spot is focused more on vertical laser shone on the tyre surface thus the circular light-spot are the ideal laser-spots. There are two steps involved in applying roundness to distinguish effective light-spots and round spot deformation degree analysis. For the depth tread of tyre measurement, this study uses Epipolar plane as shown in Figure 2.3 to measure the distance between the camera dual camera and the object

Applying Roundness to Distinguish Effective Light-spots

This study used green laser spots. By focusing the white color distribution of the laser light-spots, the binarized imaged can be extracted and the length of a_i and b_i can be found as shown in Figure 2.12. The length of a_i and b_i resemble as $r_a=a_i$, $r_b=b_i$, which defined I as small or big radius. To derive the ring's area as $B= \pi \ x \ |r_a^2-r_b^2|$, the difference between the two-concentric circle image is used while A is the area of laser light-spots failing within the ring.

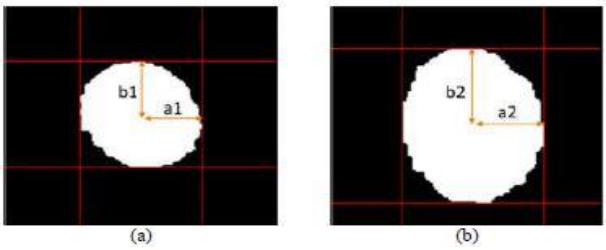


Figure 2.12: Extracted binarized image

After acquiring A and B, the property evaluation equations as shown below is used to calculate the weights of ERW_1 and ERW_2 as shown in Figure 2.13.

$$ER_1 = ERW1 \text{ X } |r_a - r_b|$$
$$ER_2 = ERW2 \text{ X } |r_a - r_b|$$

Figure 2.13: Equation ER_1 and ER_2

Round Spot Deformation Degree Analysis

This study proposed round dot deformation degree equation of ER_2 to determine the effectiveness of laser light-spot. Based on Figure 2.14, ER_1 and ER_2 are represented in error equations respectively. ER_1 is unable to differentiate between another shape effectively, while ER_2 shows a clear difference between other shape ER_2 can emphasize the shape error value more significantly compared to ER_1. Therefore, ER_2 is superior to ER_1 in differentiating the three types of shape and is used for roundness evaluation.

SHAPE	A	B	$\lvert r_a - r_b \rvert$	ER_1	ER_2
CIRCLE	124	264	1	0.46	0.53
OVAL	1245	2704	12	5.52	6.47
OTHER	567	4768	15	1.78	13.21

Figure 2.14: Light-Spot Roundness Deformation Degree

- **Tyre Tread Depth Measurement**

This study uses the Epipolar plane as shown in Figure 2.3 to calculate the relative distance between the camera and the tread of the tyre. According to Wang and Huang (2017), to obtain the laser light-spot from binarized images this study focused on two-stage grouping operations which are conventional circle Hough transform and Otsu to obtain binarized images of laser light-spot. Based on Figure 2.15, the channel Y image is extracted from the colored space by using the equation of RGB-YCbCr. The Gaussian filter is used to preserve the edge features while the Canny algorithm is used to derive edge features. Throughout the first stage of circle Hough Transform, the roundness of edge is identified and uses two times in diameter as the width of the rectangle and box ROI. In the second stage, Otsu method is used on the ROI to obtain the binarized image of laser light-spot. Morphological operations

were used to extract laser light-spot features and dynamic operations are used to calculate the laser light-spots centroid.

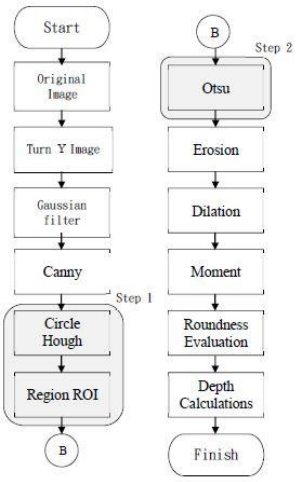


Figure 2.15: System Flow Chart

- **Experimental result**

In this study, the measurement result for depth tread of the tyre is obtained from the laser light-spot analysis and to verify the accurateness of this depth tread of tyre measurement, the depth gauge is used for depth measurement. Based on Figure 2.16, ER_2 is effective light-spot and accurately measured if the value of ER_2 is below than 7% else the measurement of depth tread of the tyre is incorrect. The measurement result of set 1-7 is effective and accurately measured compared to set 8-10.

Set	ER_1	ER_2	$Depth_{(mm)}$	$Real_{(mm)}$
1	0.68	1.31	8.073024	8.1
2	0.46	0.53	8.038665	8
3	2.94	4.05	8.004524	8
4	2.43	2.56	8.073024	8.1
5	3.54	6.45	8.004524	8
6	4.96	5.03	8.004524	8
7	3.33	5.66	7.9706	7.9
8	1.78	13.21	1.815857	8
9	2.86	9.13	9.275972	8
10	2.27	15.72	5.517561	7.9

Figure 2.16: Measurement result of depth tread of the tyre

2.4 COMPARISON OF ALL REVIEWED SYSTEM / TECHNIQUE

In Table, VI shows the comparison of four techniques to measure depth tread of the tyre. The reviewed techniques have a similar objective which to measure the depth tread of the tyre correctly by applying a different kind of techniques for instance by using light-spot analysis, image triangulation and texture analysis.

Table 2.1: Comparison of all reviewed technique

No	Author & Year	Title	Technique Proposed	Strength	Weakness	Metric	Tools
1	Tire Tech (n.d)	The conventional technique of depth tread of tyre measurement.	No technique.	Cheap tool and available in the market.	Human error in getting an accurate reading.	N/A.	Depth gauge.
2	Wang and Huang (2017)	Measuring Tread Depth with Light-spot Analysis.	Laser light-spot analysis or non-contact depth measurement.	Improve the measurement of tyre tread depth using a laser with effective analyze of light-spots.	The laser light-spots can appear differently due to the angled exposure and flatness of the exposed surface.	Light-spot roundness deformation degree such as circle and oval.	Laser. Stereo Vision.

3	Huang, Chen, and Wang (2016)	Measurement of Tire Tread Depth with Image Triangulation.	Image Triangulation.	Improve the image triangulation technique and allow the user to acquire information regarding the depth tread of tyre through the Internet.	The error rate of 0.50% or higher when pixels of left and right images are used as matching points.	Brightness level and wind size of LabVIEW.	LabVIEW stereo vision. MATLAB.
4	Huang and Syu (2014)	Texture analysis to inspect the tread worn status on the tire.	Texture Based Region of the Interested algorithm and Discriminative Robust Local Ternary Pattern.	Retain contrast information and robust to the image variation caused by the intensity inversion.	There will be a possibility the ROI failed to extract single, two, loss or incomplete major tread.	Threshold and region of interested (ROI).	N/A.

2.5 ANALYSIS OF PROPOSED TECHNIQUE AND REVIEW TECHNIQUE

This section will describe more about my proposed technique in my project with the comparison of four reviewed techniques such as the conventional technique of depth gauge, image triangulation technique, laser light-spot technique and also texture analysis of tyre.

In contrast, my proposed technique will provide more cheap and affordable computerize inspection to measure the depth tread of the tyres as my technique only need a camera and MATLAB software to process the images computerizing and give the output automatically to the users. To be compared to the review technique, image triangulation technique and laser light-spot technique required an expensive machine to perform computerize inspection on the tyre, for instance, real stereo vision environment.

Furthermore, my proposed technique allows the images of tyre taken statically by the camera without need the user to measure the depth tread of tyre manually by using the conventional technique of depth gauge. The proposed technique can accurately measure the percentage of depth tread of the tyre and give an accurate depth tread of tyre measurement to the user. The proposed technique can provide safety measurement more accurate and in an effective way, as it can alert the user to retain a tread depth of tyre in the range of standard safety measures which at least 75% of the non-interrupted band or no less than 1.6mm of tread depth.

Additionally, my proposed technique is simple and use simple algorithms to do an operation of image processing towards the tyre. The simple technique can perform the computerize inspections shortly with an accurate result of depth tread of the tyre. As the review technique involved in the more complicated machine or computerize inspection, the image processing operation towards the tyre might be longer than my proposed technique.

2.6 SUMMARY

To sum up, this chapter compares four existing technique have similar features to the proposed technique. The reviews had been done based on function, model, algorithms and experimental result and comparison of four reviewed technique had been made. Image processing technique will be implemented in this project to determine the percentage or the

status of the depth tread of the tyre. Furthermore, MATLAB software will be used as a development platform in this project

CHAPTER 3: METHODOLOGY

3.1 INTRODUCTION

This study focuses on the algorithms that will be used to determine the depth tread of the tyre. As mentioned in Chapter 1 and 2, the statistic of accidents involving cars is increasing day by day and the factors were caused by the worn tyre which has exceeded the limit of depth or the tyres are 100% worn. This situation is causing worrisome to everyone. In order to prevent this situation, there is tool provided in the market for instance depth gauge to measure the depth tread of the tyres. However, this conventional technique to measure the depth tread of the is not convenient as the computerize inspection in term of the output.

By using the conventional technique, the process to measure the depth tread of the tyre will rely on the vision. This type of inspection may not very noticeable, inefficient and unsafe to the user which has a higher chance of the misjudgment and human error in reading in order to measure the depth tread of the tyre. As compared to computerize inspection, digital image processing technique will able to inspect the tyre in an accurate and effective way. In computerization inspection, the images of the tyres will be taken by using the camera and will be analyzed by using algorithms to calculate the percentage of the depth tread of the tyre. The user does not have to inspect the tyre by themselves and just must wait for the output of the inspection.

Therefore, in order to find the suited algorithm to determine the depth tread of the tyre by using computerization inspection, few procedures should be undergoing as in methodology. This chapter will discuss more on the proposed algorithm to determine the depth tread of the tyre based on digital image processing technique. All the information and analysis throughout the Chapter 2 act as a guideline in order to find an algorithm to determine the depth tread of the tyre. The chapters will discuss in details of every stage included in the following section, for instance, requirement analysis of software and hardware requirement, methodology, process flow and conclusion.

3.2 REQUIREMENT ANALYSIS

In the requirement analysis, there are two fundamentals of requirements involved and needed to be investigated in order to assure this project is on the right track of studies. The requirement analysis includes the software requirements and hardware requirement. The software requirement usually referred to software or tool that will be used along with this project. The second requirement for carrying this study is hardware requirement. Basically, in this project, a personal computer (PC) or laptop and Digital Single-Lens Reflex (DSLR) is all needed to achieve the objective of this project.

3.2.1 SOFTWARE VERSION

MATLAB is a software which is a computer programming language or known as a proprietary programming language. MATLAB allows matrix manipulations, plotting of functions and data, implementations of various algorithms, user interface (UI) design and interfacing with other programs written in other programming languages, for instance, C, C++, C#, Java, Fortran and Python in the application. MATLAB and Simulink are also part of fundamental teaching and research tools which is founded by MathWorks. In this project, MATLAB R2015b will be the tools to implement the digital image processing algorithms in order to determine the depth tread of the tyre in term of the percentage.

3.2.2 HARDWARE REQUIREMENTS

3.2.2.1 LAPTOP

Throughout this project, in order to implement the algorithms of the image processing by using MATLAB R2015b and carrying research studies, laptop model HP Pavilion x360 Convertible was used with specifications as shown in the table below:

Table 3.1: Specification of laptop

Components	Descriptions
Laptop Model	HP Pavilion x360 Convertible
Operating System	64-bit operating system

Processor	Intel(R) Core(TM) i5-7200U CPU @ 2.50Hz 2.71GHz
Random-Access Memory	4.00 GB

3.2.2.2 DIGITAL SINGLE-LENS REFLEX (DSLR)

In this project, in order to collect data or images of new and used tyres, digital camera or DSLR model Canon EOS 50D camera was used with specifications as shown in the table below:

Table 3.2: Specification of the camera

Components	Description
Model	Canon EOS 50D
Image processor	DIGIC 4
Sensitivity	ISO 100 to 3200, extendable to H1 6400 to H2 12800
Noise reduction	Four levels
Auto light optimizer	Four levels

3.3 METHODOLOGY

According to William (2014), the methodology can be delineating as an approach used to give a clear concept of what the researcher is investigating in their research or project. The methodology can be the right platform for the researcher to map out their research work in relevance to make a solid plan. In addition, the methodology can also be a guide to the research to involve and be active in the field of study.

The entire research plan is based on the concept of research framework or methodology. Furthermore, methodology might assist the researcher by providing an idea of setting up the right research objective, followed by literature points of view, based on the chosen analysis through any approaches used as the guidance.

On this project, qualitative research method is used to obtain the objective of this study. Qualitative research is a scientific research that was used to seek an answer off to solve the problem statements with a set of procedure. For this project, there are 5 stages of the

methodology involved in order to determine the depth tread of the tyre. The stages involved are research question and background, data collection, research experimentation, result and conclusion and lastly is report writing.

The figure below illustrates the process flow or research framework to determine the depth tread of the tyre. Every stage will be discussed further in the following section.

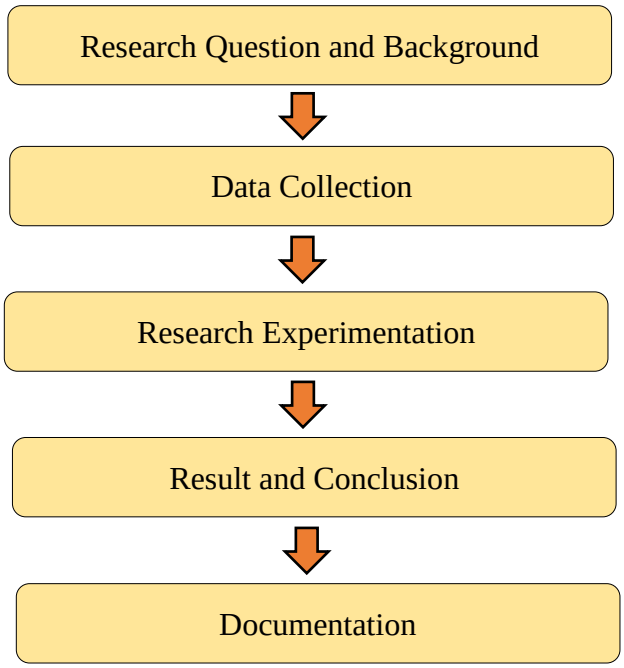


Figure 3.1: Process flow of the methodology

3.3.1 PHASE 1: RESEARCH QUESTION AND BACKGROUND

In this phase, the research question begins with a research problem of an issue of the situation that needs to be changed or addressed. In this research-based, the research question will be what algorithm in digital image processing that can be used to measure the depth tread of tyre measurement as well to give an accurate percentage of depth tread of the tyre. Background of this project is also based on a literature review in Chapter 2. It is required to explore the details of the previous reference of studies in orders to enhance understanding on research, provides clues and allows a better understanding of digital image processing field.

3.3.2 PHASE 2: DATA COLLECTION

Data collection is the second step involved in the methodology to collect the sample images of the car tyres. The image is captured in a controlled environment and the process of capturing the image is shown in Figure 3.2 and Figure 3.3.

The steps involved to capture the images of the tyre under controlled environment are:

1) Turning the steering wheel to one side that will give a better view of the tread surface on the front tyres as shown in Figure 3.2 and Figure 3.3.

2) Set a constant distance between the camera and the tyres.

3) Set the digital camera settings and mode as follows:

- Shutter speed of the camera is set to 1/60;
- The aperture of camera setting is set to F5.6;
- ISO of camera setting is set to 3200;

4) Capture the images.

5) Repeat the step for the new tyres.

Figure 3.2: Top view of setup

Figure 3.3: Side view of setup

By capturing the images of the tyres under the controlled environment, it helps to reduce the amount of unwanted information such as noise. In addition, it helps to ensure that the images are clear, sharp and appropriate to undergo image processing.

In this phase, all the data for the research should be collected before research experimentation phase. Therefore, the data is collected by capturing five different images of tyre tread as training dataset. The tyres tread image is taken from the different cars and groove by using DSLR (Digital single-lens reflex). There will be ten images of tyre tread which are five images of the new tyre and five images of the used tyre. During this process, the car registration plate will be recorded. Some of the sample images are shown in Figure 3.4 to Figure 3.5.

The images of the new and used tyre will be converted into JPEG (Joint Photographic Expert Group) format if the images are not in the desired format in the first place. The purpose of converting the images into JPEG format is to achieve standardization process. JPEG is a typical file format of the most digital camera and it is the best used in the still images. JPEG also has a small size and capability to reduce the processing time while improving the efficiency. The image resolution will set to 1100 x 1100 pixel. Few conversions will be performed by converting the RGB (Red, Green, Blue) into greyscale images. Some of data sample collection is represented in Table 3.3.

Figure 3.4: Tread of the new tyres captured under controlled environment

Figure 3.5: Tread of the used tyres captured under controlled environment

Table 3.3: Some of Data Collection for Tyres

No	New Tyres	Used Tyres
1		
2		

3.3.3 PHASE 3: RESEARCH EXPERIMENTATION

After data collection in phase 2 is complete, research experimentation will be carried out in order to implement the three chosen algorithms and determine the most suited algorithm to measure the depth tread of the tyre by using computerization inspection. The experiment will be conducted between the new and the used car tyres. In this project, the image of the new and worn tyres will be compared with one another by using three algorithms which are Edge detection, Histogram algorithm and Hausdorff Distance algorithm. The purpose of using three different algorithms is to find the most accurate algorithm between these three algorithms which will able to give the desired result. The process flow in Figure 3.6 shows the process of determining the depth tread of the tyre by using image processing.

The process flow for measuring the depth tread of the tyre:

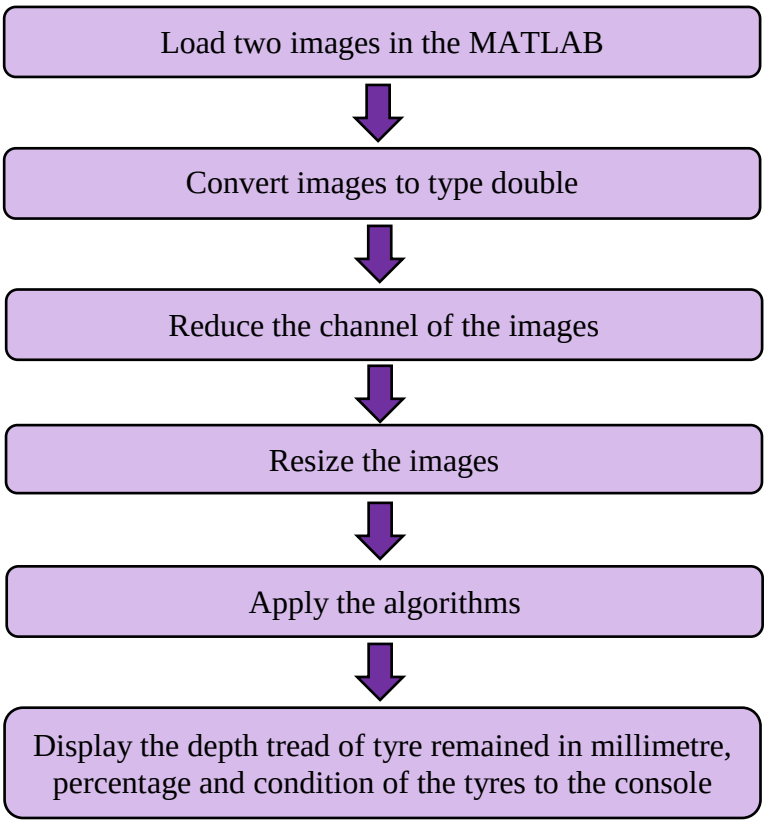


Figure 3.6: Process flow of proposed technique

The algorithms for measuring depth tread of the tyres begin by loading two images of the new and worn tyres in the MATLAB by using "imread" command. This command infers from the file that the graphics file format is JPEG (Joint Photographic Expert Group).

After that, the images are converted to type double which is range from 0 to 1, instead of from 0 to 255 by using "im2double" command. This command converts the intensity of the image I to the double precision and rescaling the data if the necessary this is because when operations on images are easier when the images are represented in floating point. For example, when comparing one image from another, if the images are in unsigned integer format, everywhere the second image was greater than the first would get 0 as the result because unsigned integers cannot represent negative numbers.

The channel of the images is also reduced from three channel RGB (Red, Green, Blue) into one channel grayscale by using "rgb2gray" command. This command converts the truecolor

RGB image or colormap to grayscale by eliminating the hue and saturation information while retaining the luminance. The image is converted from RGB format to grayscale format because if there are any unnecessary color present on the image beside the tyres, it will effect on the result when the image of new and worn tyres is compared.

The images are also resized to 1100 x 1100 pixels by using “imresize” command, followed by applying the algorithms. The image is resized into the same pixels so that all the image will achieve standardization in size.

The result of depth tread of the tyres remained in millimeter (mm), percentage worn and condition of the tyres is displayed to the console.

PROPOSED ALGORITHM

The priority of this research-based project is to determine the algorithms that can be used to measure the percentage of depth tread of the tyre by using digital image processing. In this project, image comparison using Histogram, Edge detection and Hausdorff Distance will be used to find the depth tread of the tyres remained by using computerized inspection. These three algorithms will be testing and compared in term of accuracy to find the finest algorithm to measure the depth tread of the tyre that can give the desired result. The three algorithms will be following the process flow in Figure 3.6. This section will explain in detail about the three algorithms that will be used.

Algorithm 1: Image Comparison Using Histogram

According to Nguyen (2016), there are many image comparison algorithms that can be used to compare two images which are Short Time Fourier Transform (STFT), wavelet, edge detection and also histogram.

The histogram can be elaborated as a way of visualizing the largest intensities of an image and used to describe problems which begin in image acquisition such as exposure, contrast and dynamic range. The contrast of grayscale image shows how the object in the image can be differentiated. The dynamic range is the number of the distinct pixel value in an image. The histogram is also a graph used to show the number of pixels in an image at each different

intensity value found in the image. The histogram of the digital image is a distribution of its discrete intensity level in the range [0, L−1].

Image comparison using histogram used the images of new and worn tyres to find the difference by measuring the histogram error between the images. The technique relies on finding the normalized histogram of both images of the tyres and calculate the sum of squared difference between the normalized histogram. The normalized histogram plots the number of pixels for each tonal value of the images.

The histogram can be calculated by using the function:

$$\mathrm{h(v)} = \mathrm{round}\ (\frac{cdf(v) - cdf_{min}}{(M \times N) - cdf_{min}} \times (\mathrm{L} - 1))$$

where

cdf_{min}: minimum value of the cumulative distribution function;

$M \times N$: number of pixels where M and N is width and height respectively;

L: number of gray levels used.

The two images are divided into rows and column and used the histogram function to calculate the histogram value between each point of the images to achieve two different histogram values for the same point which shows the difference between two images. The normalized value becomes by completing the calculation of the histogram values.

After algorithm is applied, the histogram graph is plot and the result of the depth tread of the tyres remained is displayed in the console as well as the percentage worn and the condition of the tyres.

Algorithm 2: Image Comparison Using Edge Detection

According to Ehsan and Haddadnia (2011), edge detection is one of the commonly used technique to locate the pixels in the image that coincide to the edge of the objects seen in the image in digital image processing. Edge detection is the name for a set of the mathematical technique used to identify points in a digital image at which the image brightness changes or

has discontinuities or noise. The points at which image brightness alters sharply and typically organized into a set of curved line segments termed edges.

There are various of the various edge-detection method which are Sobel edge detection, Perwitt edge detection, Roberts edge detection, Zerocross threshold edge detection and also Canny edge detection. In this project, to compare the image of new and worn tyres, the technique that will be used is Sobel edge detection techniques.

According to Bin and Yeganeh (2012), Sobel operator is in the form of filtering operator and it is used to extract the edge of the object in the digital images. Each point in the images are the two nuclear loops, which one checks the maximum response of the vertical edge and the one check the maximum response of the horizontal edge. The maximum value of two loops will be referred as the output value of the changing point.

The Sobel operator as shown in Figure 3.7 is easy to achieve in space which has a smoothing effect on the noise that nearly affected by noise as well as provides more accurate edge direction information, but it will also detect many false edges with coarse edge width.

-1	-2	-1
0	0	0
1	2	1

-1	0	1
-2	0	2
-1	0	1

Figure 3.7: Sobel Operator

Algorithm 3: Image Comparison Using Hausdorff Distance

According to Huttenlocher, Klanderman, and Rucklidge (1993), Hausdorff Distance can be calculated by measuring the extent of every point appear on a model that agreed on some point of a related image set or vice versa. Hence, this distance can be used to determine the degree of correspondence between the image of two objects. The Hausdorff Distance is measured based on the point sets and not for segments and Hausdorff Distance is a metric

over the set of all closed and bounded sets. The minimal Hausdorff Distance for grid points compute the distance transform and is set as a function of translation. Image comparison using Hausdorff Distance compare the portion of shapes and models of the image.

3.3.4 PHASE 4: RESULT AND CONCLUSION

Experiment in phase 3 is carried out to determine the depth tread of the tyre between new and worn tyre. The result of depth tread of the tyre from three algorithms in research experimentation will be compared with Table 3.4 to check the condition of the tyre either the condition of the tyre is still in good condition or have reached the legal limit. Furthermore, to determine which algorithm to choose, the results of the three algorithms in research experimentation will be listed in the table and be compared which each other. The result that indicated the highest percentage will be chosen as the algorithm to determine the depth tread of the tyre in digital image processing.

In this research-based project, five images of the new and worn tread of the tyre are taken as a training dataset to determine the depth tread of the tyre. The percentage of tyre worn indicates the remaining tyre depth in millimeter (mm). To decrease the numbers of accidents which caused by worn tyres, every car must follow the legal measurement of the tyre tread depth which greater than 1.6 mm. The tyre tread measurement technique by using digital image processing can accurately measure the percentage of depth tread of the tyre and alert the user to retain a tread depth of tyre in the range of standard safety measure which at least 75% of the non-interrupted band or no less than 1.6mm of tread depth. The percentage of tyre worn is presented in the table form as shown in below:

Table 3.4: The list of remaining tyre depth, percentage, and the tyre conditions

Remaining tyre depth	Percentage of tyre worn	Condition
8mm	Tyres are 0% worn	Good
7mm	Tyres are 15% worn	
6mm	Tyres are 31% worn	
5mm	Tyres are 47% worn	
4mm	Tyres are 62% worn	
3mm	Tyres are 78% worn	Inspect Monthly
2mm	Tyres are 94% worn	Replace urgently

1.6mm	Tyres are 100% worn	Legal limit

3.3.5 PHASE 5: DOCUMENTATION

In the last phase of methodology, documentation is required to present the information and give details after the research has been carried out. The report will also recommend the proposed algorithm that can be used to measure depth tread of car tyres by using digital image processing and give the full information about the project in details.

3.4 CONCLUSION

In conclusion, the proposed algorithm for measuring depth tread of the tyres by using digital image processing are discussed in this chapter. This chapter also discusses the requirement analysis of software version and hardware requirement, methodology, process flow and conclusion.

CHAPTER 4: REQUIREMENT ANALYSIS AND DESIGN

4.1 INTRODUCTION

Digital image processing is an interesting field in computer science as it gives a bettered pictorial information for human interpretation and processing of digital image data for storage, transmission, and depiction for machine perception. Image processing is a technique to enhance raw image received from camera or sensors placed on cars, satellites or pictures taken in normal day-to-day life for an individual purpose. The field of digital image processing is undoubtedly improved in recent times and extended to various fields of science and technology. The image processing mainly pledges with image acquisition, image enhancement, feature extraction, image classification, image comparison and others.

This research-based project of Tyre Tread Wear Indicator Using Digital Image Processing applies the image comparison between new and worn tyres to achieve the main objective of this project which is to measure depth tread of the tyres. Image comparison is chosen because it is more basics technique in image processing, which can measure the depth tread of tyres in a short time, involved mathematical calculation and applicable to this research-based project as in this project, the image of new and worn tyres are captured from the same tread and to measure the depth tread of the tyres, it can simply be measured by comparing both images by using algorithms.

In this chapter, the algorithms to measure depth tread of the tyres has been implemented based on the whole requirements in the objective and Chapter 3. All the entire data has been collected during data collection process were fully used in order to develop a good algorithm. Therefore, this phase involved the implementation of the three algorithms that can be used to measure depth tread of the tyres. To meet the objective of the project, GUIDE (Graphical User Interface Development Environment) with MATLAB is implemented to ease the measurement of depth tread of tyres by comparing new and worn tyres. This chapter will start by present the dataset that is being used to test the algorithms, followed by present the algorithms to measure depth tread of the tyres, GUIDE, results, and conclusion of the chapter.

4.2 DATASET

Five images of the tyres are collected during data collection process. The image is converted into JPEG (Joint Photographic Expert Group) format to achieve standardization process. Table 4.1 shown the dataset that has been collected. In order to attain uniformity between the tyres, the tyres between the new and worn tyres must have the same type of groove, tread and model. The tyres are rename according to the file name as shown in Table 4.1 to ease recognition of which tyres are testing during research experimentation in MATLAB.

Table 4.1: Dataset of Tyres and File Name

Tyre No	**File Name**	
Tyres 1	imagenew1.jpg	imageused1.jpg
Tyres 2	imagenew2.jpg	imageused2.jpg
Tyres 3	imagenew3.jpg	imageused3.jpg

Tyres 4	 imageused4.jpg	 imageused4.jpg
Tyres 5	 imagenew5.jpg	 imageused5.jpg

4.3 TECHNIQUE FOR COMPARING TWO IMAGES

The technique for comparing two images is constructed based on process flow in Figure 3.6.

Step 1: Reading the images.

The two images of tyres are reading in the workspace by using *imread* and the image is in JPEG format to achieve standardization.

```
%% Read two images
ImageNew = imread('imagenew1.jpg'); % Image of the new tyres
ImageUsed = imread('imageused1.jpg'); % Image of the used tyres
```

Step 2: Converting the images into double type.

The images of new and worn tyres is converted into double type by using im2double and the image is converted from range 0 to 1 instead of 0 to 255.

```
%% Convert images to type double
ImageDoubleNew = im2double(ImageNew);
ImagedDoubleUsed = im2double(ImageUsed);
```

Step 3: Reducing the channel.

The images are reduced from three channel (RGB) into one channel (Grayscale) by using *rgb2gray*.

```
%% Reduce three channel (RGB) to one channel (grayscale)
ImageGrayNew = rgb2gray(ImageDoubleNew);
ImagegGrayUsed = rgb2gray(ImagedDoubleUsed);
```

Step 4: Resizing the image.

The images are resized to 1100 pixels x 1100 pixels to achieve standardization in size by using *imresize*.

```
%% Resize the image to [1100 1100]
ImageGrayNew1=imresize(ImageGrayNew,[1100 1100]);
ImageGrayUsed1=imresize(ImagegGrayUsed,[1100 1100]);
```

Step 5: Applying the algorithm to find the difference between new and worn tyres.

After applying the step 1 until step 4 in the first part of programming, the algorithm of Histogram is applied for algorithm 1, an algorithm for Edge detection is applied for algorithm 2 and Hausdorff Distance is applied for algorithm 3.

Step 6: Display result to the console.

The percentage of depth tread of the tyres is displayed to the console based on the depth calculated in Step 5.

```
%% Display the result to the console
fprintf('The remaining depth tread of tyre is : %.4f mm.\n', depth);

%% Display in percentage
if depth <= 1.6
    fprintf('The percentage of tyre is 100 worn! Reached legal limit.\n');
elseif 1.6 < depth && depth <=2
    fprintf('The percentage of tyre is 94 worn! Replace Urgently.\n');
elseif 2 < depth && depth <=3
```

```
    fprintf('The percentage of tyre is 78 worn! Inspect Monthly.\n');
elseif 3 < depth && depth <=4
    fprintf('The percentage of tyre is 62 worn! Tyre is good
condition.\n');
elseif 4 < depth && depth <=5
    fprintf('The percentage of tyre is 47 worn! Tyre is good
condition.\n');
elseif 5 < depth && depth <=6
    fprintf('The percentage of tyre is 31 worn! Tyre is good
condition.\n');
elseif 6 < depth && depth <=7
    fprintf('The percentage of tyre is 15 worn! Tyre is good
condition.\n');
elseif depth >= 7 && depth <=8
    fprintf('The percentage of tyre is 0 worn! Tyre is good condition.\n');
else
    fprintf('Error has occured.\n');
end
```

4.4 ALGORITHMS

To refine this study, the priority of this research-based project is to determine the algorithms that can be used to measure the percentage of the depth tread of the tyres. Therefore, in this project, three algorithms will be implemented and compared in term of accuracy to find the finest algorithms to measure the depth tread of the tyres. According to Nguyen (2016), there are many image comparison algorithms that can be used to compare two images which are Short Time Fourier Transform (STFT), wavelet, edge detection, and histogram. As a result, in this research-based project Histogram, Edge Detection and Hausdorff Distance are the three algorithms that have been chosen to undergo research experimentation. After the implementation of algorithms, the result of each algorithm is presented and the result for each algorithm is compared to find the finest algorithms that will be used to measure the depth tread of tyres.

4.4.1 ALGORITHM 1: IMAGE COMPARISON USING HISTOGRAM

To recall back, the histogram is a way of visualizing the largest intensities of an image and it is used to depict problems which originate during image acquisition. The contrast of a grayscale image that are converted by using *rgb2gray* in Step 3 shows how easily object in the image can be distinguished. The dynamic range is the number of the distinct pixel value in an image and it is used to show image statistics in an easily interpreted visual format. Based on this algorithm, the graph showing the number of pixels in an image at each different intensity value found in that image is plotted along with the result. The histogram of the digital images of the tyres is a distribution of its discrete intensity levels in the range [0, L-1].

The operation involved in comparing the images of new and worn tyres is simple. The images are scanned in a single pass and a running count of the number of pixels found at each intensity value is kept. Then, it is used to construct the normalized histogram of new and worn tyres using *imhist* and *numel* in MATLAB. The *imhist* is used to calculate the histogram for the intensity image of the tyres and display a plot of the histogram. The number of the bin in the histogram is determined by the image type and the images of the tyres used in this project are a Grayscale image, therefore, *imhist* uses 256 bins by default. The *numel* is using to calculate the array of size L=256.

```
%% ALGORITHM 1 | FOR FINAL YEAR PROJECT 2

%% Calculate the Normalized Histogram of New Tyres and Used Tyres
NormalizedHistogram1 = imhist(ImageGrayNew1)./numel(ImageGrayNew1);
NormalizedHistogram2 = imhist(ImageGrayUsed1)./numel(ImageGrayUsed1);
subplot(2,2,1);subimage(ImageGrayNew1)
subplot(2,2,2);subimage(ImageGrayUsed1)
subplot(2,2,3);plot(NormalizedHistogram1)
subplot(2,2,4);plot(NormalizedHistogram2)
```

In order to compare the images between the new and worn tyres, histogram error is used after the histogram is constructed to get the difference between the two images. As the depth tread of the tyres is in millimeter (mm), to achieve it the result of depth tread of the tyres in mm, the histogram error is multiplied with 1000.

```
%% Calculate the histogram error
HistogramError = sum((NormalizedHistogram1 - NormalizedHistogram2).^2);
```

```
%Sum of Difference
depth = HistogramError *1000; %Converting to millimeter
```

A grayscale image is usually encoded with n=8 bits and an array of size L=256. The number of possible intensity values is starting from 0 and 255 where 0 is black and 255 is white. Since there are 256 different possible intensities, the histogram will graphically display 256 numbers showing the distribution of pixels among the grayscale value. When Histogram is plotted, x-axis represents the intensity value from 0 to 255 while y-axis represents the number of pixels with the intensity values. The y-axis will vary depending on the number of the pixels in the image and how their intensities are distributed.

The following figure illustrates the difference between the displays of the histogram of the image of new and worn tyres. The image on the left is the image of new tyres and the image on the right is the image of worn tyres. Figure 4.1, 4.2, 4.3, 4.4 and 4.4 shows the MATLAB result for comparing two images of new and worn tyres. The histogram result of all figure exhibits a good bi-modal distribution. Based on the histogram plotted one peak represents the object pixels, other represent the background of the images.

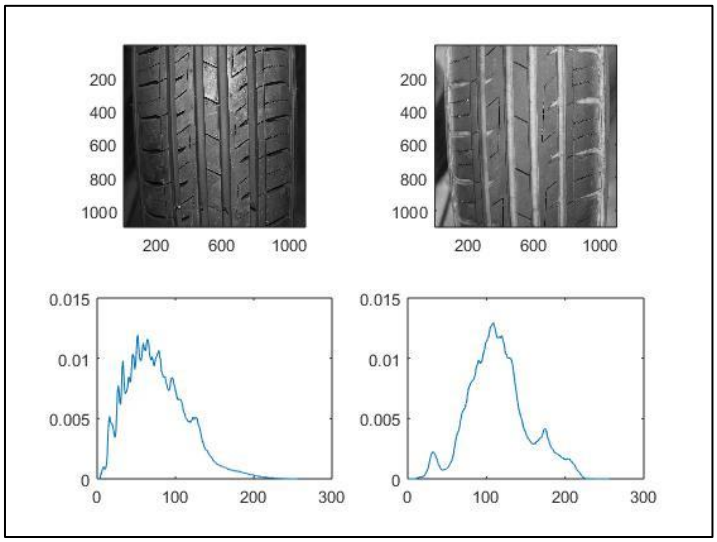


Figure 4.1: Result of Tyres 1

The resulting of the histogram as shown in Figure 4.1 showed a large spike at 52 intensity value of new tyres and 105 intensity value of worn tyres which results in the rest of the histogram being scaled up to 256.

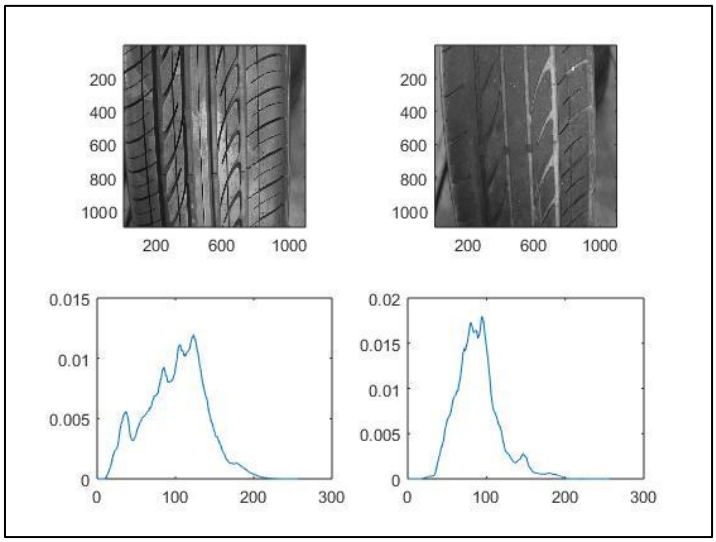


Figure 4.2: Result of Tyres 2

The resulting of the histogram as shown in Figure 4.2 showed a large spike at 123 intensity value of new tyres and 94 intensity value of worn tyres which results in the rest of the histogram being scaled up to 256.

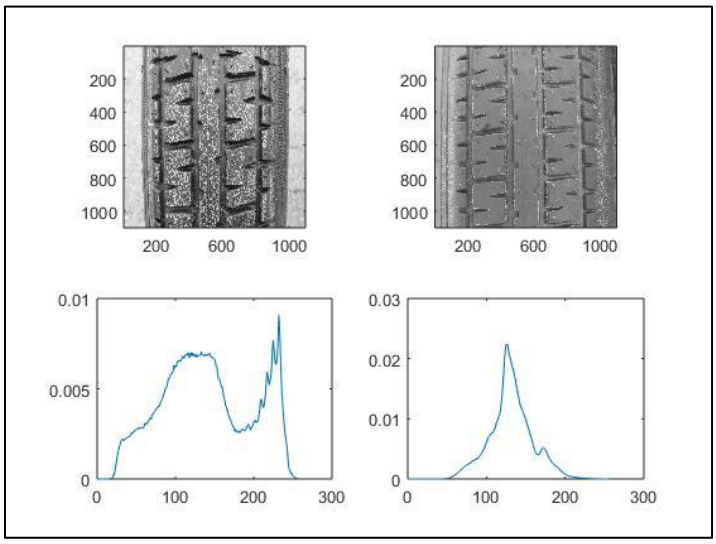


Figure 4.3: Result of Tyres 3

The resulting of the histogram as shown in Figure 4.3 showed a large spike at 232 intensity value of new tyres and 152 intensity value of worn tyres which results in the rest of the histogram being scaled up to 256.

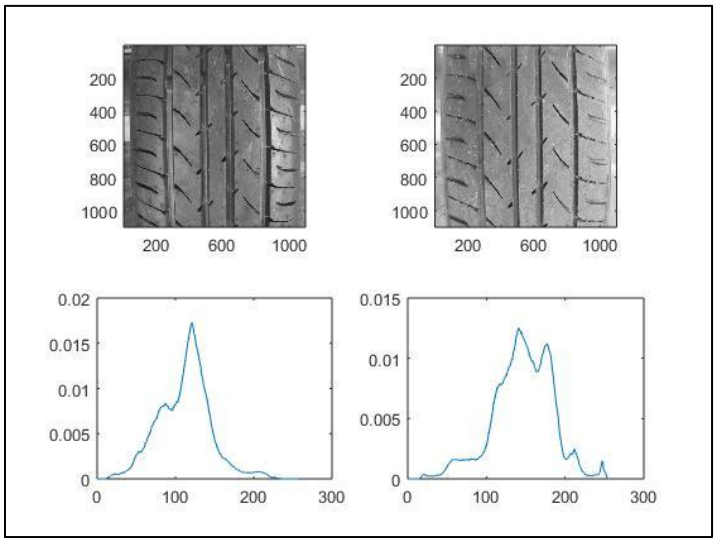


Figure 4.4: Result of Tyres 4

The resulting of the histogram as shown in Figure 4.4 showed a large spike at 121 intensity value of new tyres and 141 intensity value of worn tyres which results in the rest of the histogram being scaled up to 256.

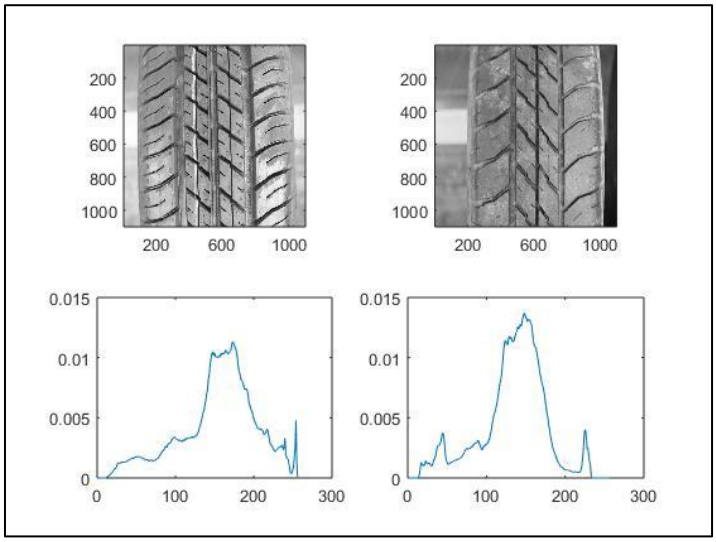


Figure 4.5: Result of Tyres 5

The resulting of the histogram as shown in Figure 4.5 showed a large spike at 173 intensity value of new tyres and 149 intensity value of worn tyres which results in the rest of the histogram being scaled up to 256.

To summarize, based on the resulting of the histogram, the intensity value at the large spike in the image of new and worn tyres is obtained by using *imhist.* After the intensity value is obtained, the normalized histogram of the image of new and worn tyres is constructed by dividing the intensity value obtained with the array by using *numel.* In order to obtain depth tread of the tyres by using image comparison of the histogram, error histogram is calculated based on the result of the normalized histogram.

4.4.2 ALGORITHM 2: IMAGE COMPARISON USING EDGE DETECTION

To recall back, edge detection is one of the commonly used technique to locate the pixels in the image that coincide with the edge of the objects seen in the image in digital image processing. Edge detection is the name for a set of the mathematical technique used to identify points in a digital image at which the image brightness changes or has discontinuities or noise. The points at which image brightness alters sharply and typically organized into a set of curved line segments termed edges.

Sobel operator is in the form of filtering operator and it is used to extract the edge of the object in the digital images. Each point in the images are the two nuclear loops, which one checks the maximum response of the vertical edge and the one check the maximum response of the horizontal edge. The maximum value of two loops will be referred as the output value of the changing point.

The operation involved in comparing the images of new and worn tyres is simple by using edge detection. After the image of the new and worn tyre is resized, the edge detection is applied to the image of the new and worn tyres in order to obtain the point in the images of tyres.

```
%% ALGORITHM 2 | FOR FINAL YEAR PROJECT 2

%% Edge detection is applied on new and worn tyres to obtain white and
black points as well as the edge presents in the image
EdgeDetNew1 = edge(ImageGrayNew1,'sobel'); % Edge detection is applied on
the image of new tyre
EdgeDetUsed1 = edge(ImageGrayUsed1,'sobel'); % Edge detection is applied on
the image of worn tyre
```

```
EdgePoints = 0; EdgePoints1 =0; WhitePoints = 0; BlackPoints = 0;
```

The for loop is applied to the image of new tyres to obtain the black point as well as the white point on the images of new and worn tyres. After that, for loop is used to compare the edge points in the image of new and worn tyres.

```
%% For loop used for detecting black and white points in the images
 for a = 1:1:1100
    for b = 1:1:1100
        if(EdgeDetNew1(a,b)==1)
            WhitePoints = WhitePoints+1;
        else
            BlackPoints = BlackPoints+1;
        end
    end
end

%% For loop comparing the edge points in the two images
 for i = 1:1:1100
    for j = 1:1:1100
        if(EdgeDetNew1(i,j)==1)&&(EdgeDetUsed1(i,j)==1)
            EdgePoints = EdgePoints+1;
        else

            EdgePoints1 = EdgePoints1+1;
        end
    end
end

%% Edge detection of new and worn tyres
subplot(2,2,1);subimage(ImageGrayNew1)
subplot(2,2,2);subimage(ImageGrayUsed1)
subplot(2,2,3);subimage(EdgeDetNew1)
subplot(2,2,4);subimage(EdgeDetUsed1)
```

From the for loop, the total data of white points and total difference resulting of comparing the edge point on the image of new and worn tyres is calculated by diving the edge point with total data of white points obtained. As the depth tread of the tyres is in millimeter (mm), to

achieve it the result of depth tread of the tyres in mm, the histogram error is multiplied with 100.

```
%% Display the result to console
fprintf('The matched data is: %.f. \n',EdgePoints);
fprintf('The total data is: %.f. \n',TotalData);
fprintf('The remaining tyre tread depth is : %.4f mm.\n', depth);
```

The following figure illustrates the difference between black and white point on the image of the new and worn tyre when edge detection is applied. The total data of white points and the corresponding data is calculated at the end of each iteration of for loop. The image on the left is the image of the new tyres and the image on the right is the image of the worn tyres.

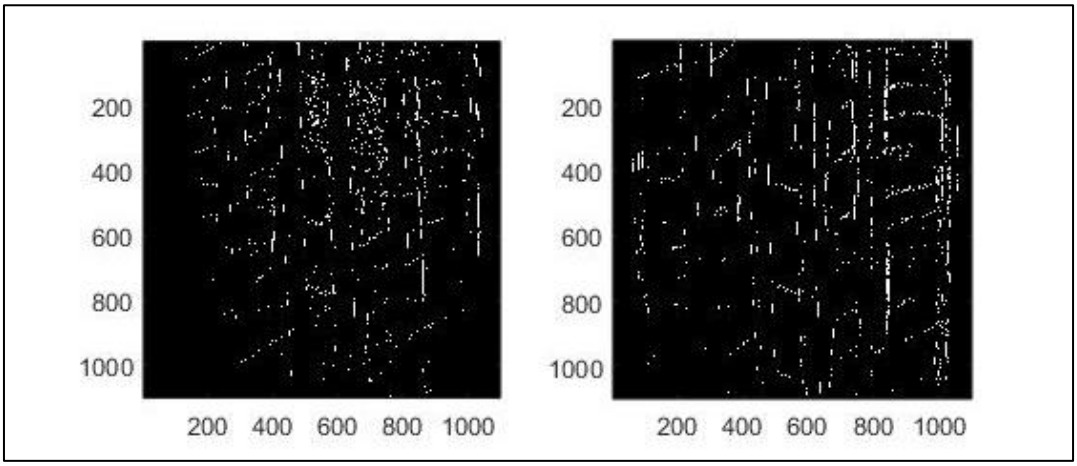


Figure 4.6: Result of Tyres 1

The resulting of the edge detection as shown in Figure 4.6 showed the total data of the white point of both images of new and worn tyres is 31955 and total difference of corresponding data is 1460 points.

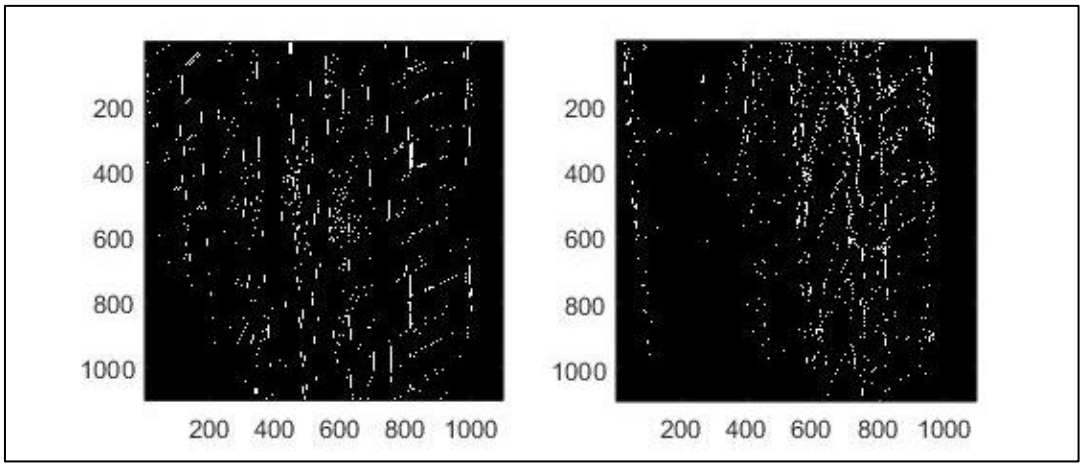


Figure 4.7: Result of Tyres 2

The resulting of the edge detection as shown in Figure 4.7 showed the total data of the white point of both images of new and worn tyres is 34800 and total difference of corresponding data is 1492 points.

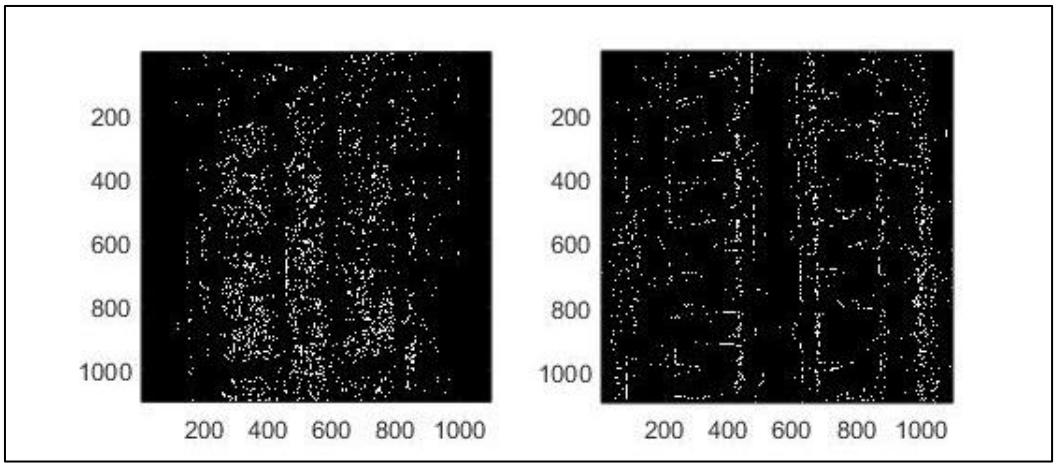


Figure 4.8: Result of Tyres 3

The resulting of the edge detection as shown in Figure 4.8 showed the total data of the white point of both images of new and worn tyres is 59798 and total difference of corresponding data is 1820 points.

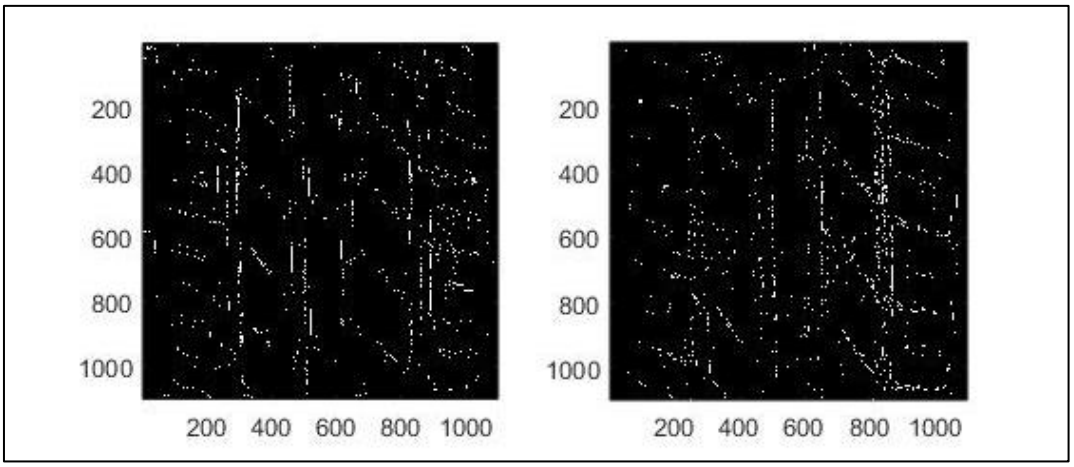


Figure 4.9: Result of Tyres 4

The resulting of the edge detection as shown in Figure 4.9 showed the total data of the white point of both images of new and worn tyres is 29079 and total difference of corresponding data is 1191 points.

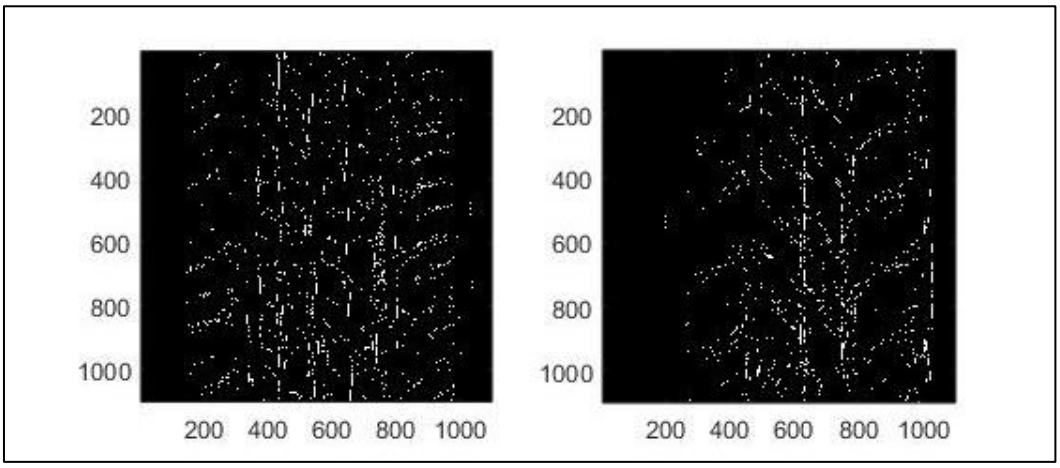


Figure 4.10: Result of Tyres 5

The resulting of the edge detection as shown in Figure 4.10 showed the total data of the white point of both images of new and worn tyres is 35803 and total difference of corresponding data is 1137 points.

To summarize, in order to measure depth tread of the tyre by using edge detection of soble, the algorithm first applying the edge detection on the image of the new and worn tyres. After that, a total of the white and black point is obtained on the image of new tyres by using for

loop. The process is followed by comparing edge point on the image of new and worn tyres by using for loop. Next, after the total of white total data of white point and edge points is calculated based on the edge detection, the measurement of depth tread of tyre by using comparison of edge detection between the image of new and worn tyres is calculated by multiplier 100 with the total difference of edge point over total data to get the result in millimetre.

4.4.3 ALGORITHM 3: IMAGE COMPARISON USING HAUSDORFF DISTANCE

To recall back, Hausdorff Distance can be calculated by measuring the extent of every point appears on a model that agreed on some point of a related image set or vice versa. Hence, the distance can be used to determine the degree of correspondence between the image of two objects. The Hausdorff Distance is measured based on the point sets and not for segments and Hausdorff Distance is a metric over the set of all closed and bounded sets. The minimal Hausdorff Distance for grid points compute the distance transform and is set as a function of translation. Image comparison using Hausdorff Distance compare the portion of shapes and models of the image

The operation involved in comparing the images of new and worn tyres is simple. The Hausdorff distance between the image of the new and worn tyre is denoted by dH (ImageGrayNewA, ImageGrayNewB) and is defined by dH = max (compute_distance (ImageGrayNewA, ImageGrayNewB)) where ImageGrayNewA is the image of new tyre and ImageGrayNewB is the image of a worn tyre. Size of the ImageGrayNew1 and ImageGrayUsed returns a row vector whose element contain the length of the equivalent dimension of ImageGrayNew1 and ImageGrayUsed1. The measurement of depth tread of tyre can be achieved after the distance is compute by multiplying with 10.

```
%% ALGORITHM 3 | FOR FINAL YEAR PROJECT 2

 function dH = hausdorff( ImageGrayNewA, ImageGrayUsedB)

%% Compute distance of an object in image of new and worn tyres
ImageGrayNewA = ImageGrayNewA(ImageGrayNewA >= 0);
ImageGrayUsedB = ImageGrayUsedB(ImageGrayUsedB >= 0);
dH = max(ComputeDistance(ImageGrayNewA, ImageGrayUsedB),
```

```
ComputeDistance(ImageGrayUsedB, ImageGrayNewA)); % Calculate the maximum
distance

%% Plot
    subplot(2,2,1);subimage(ImageGrayNew)
    subplot(2,2,2);subimage(ImagegGrayUsed)
end
```

Size function is used on the image of new tyres as well as worn tyres to return a row vector whose elements contain the length of the corresponding dimension of the new and worn image of the tyres.

```
%% Compute distance
function dists = ComputeDistance(ImageGrayNew1, ImageGrayUsed1)

ImageGrayNewM = size(ImageGrayNew1); % Returns a row vector whose elements
contain the length of the corresponding dimension of ImageGrayNew1
ImageGrayUsedN = size(ImageGrayUsed1); % returns a row vector whose
elements contain the length of the corresponding dimension of
ImageGrayUsed1

distance = []; D = [];
```

After the length of the corresponding dimension of the new and worn image of the tyres is get, for loop is used to loop through the all point looking for the absolute value while returns the length of dimension by comparing the image of new tyres and worn tyres and plot the result of the algorithm.

```
%% Loop through all points looking for absolute value
for j = 1:ImageGrayNewM(2) % Returns the length of dimension 2
    for k= 1: ImageGrayUsedN(2) % Returns the length of dimension 2
        D(k) = abs((ImageGrayNew1(j)-ImageGrayUsed1(k))); % Absolute value
    end
end

%% Plot
    subplot(2,2,3);plot(ImageGrayNew1)
    subplot(2,2,4);plot(ImageGrayUsed1)
```

Measuring the depth tread of the tyres by using the absolute value calculated.

```
%% Measure depth tread of the tyre
distance(j) = min(D);
dists =sum(max(distance).*2);
depth = dists *10;
```

The following figure contain the length of the corresponding dimension between the new and the worn tyres that will be used to obtain the absolute value and be used to measure the depth tread of the tyre. The image on the left is the image of the new tyres and the image on the right is the image of the worn tyres.

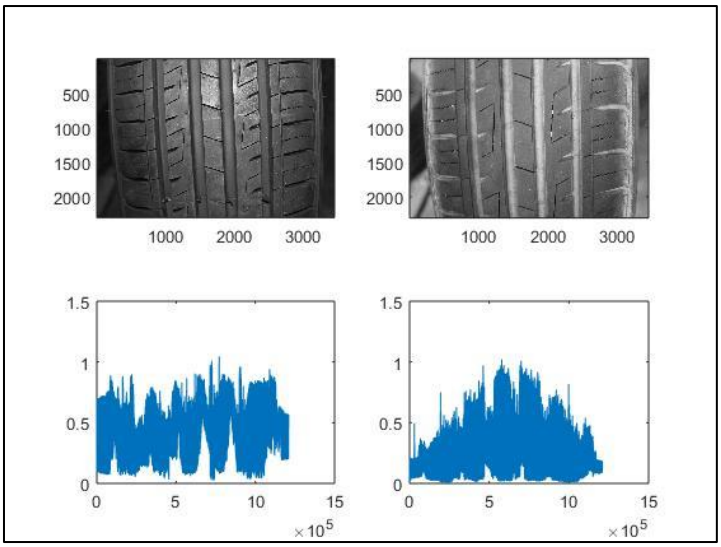


Figure 4.11: Result of Tyres 1

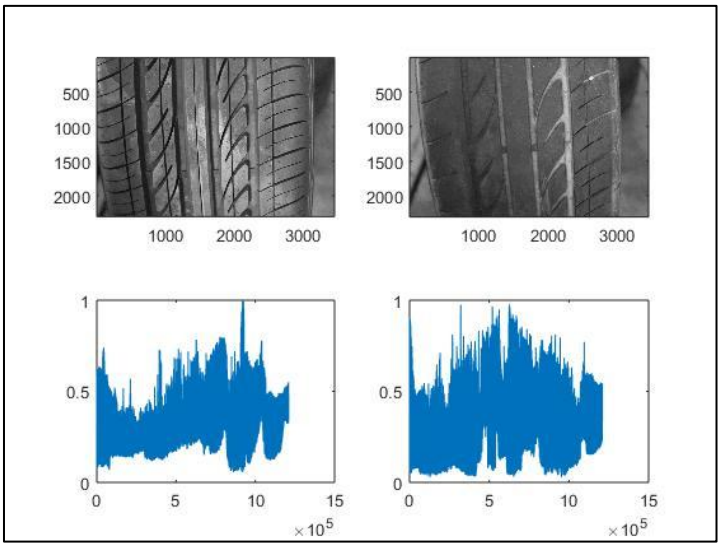


Figure 4.12: Result of Tyres 2

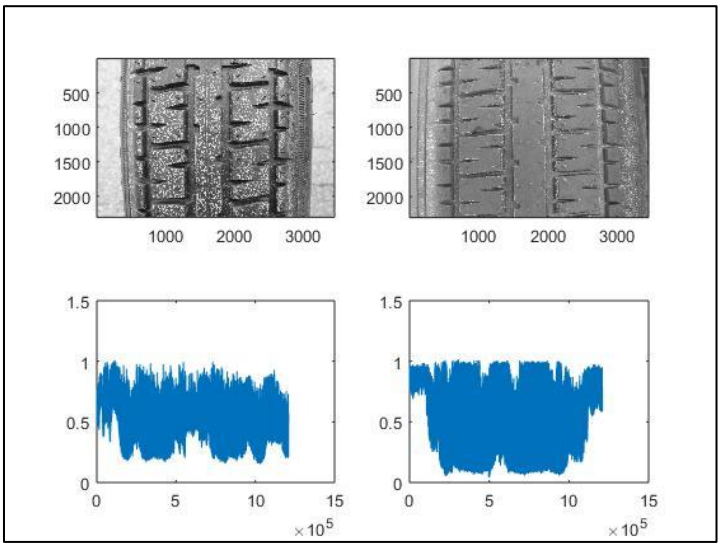


Figure 4.13: Result of Tyres 3

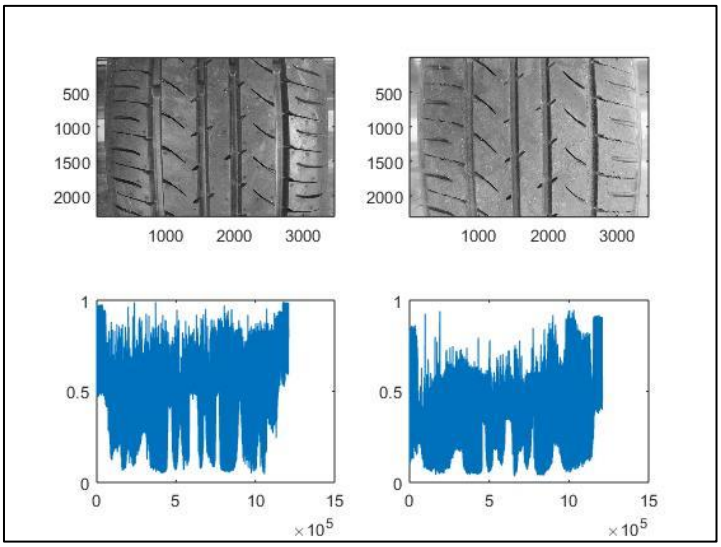


Figure 4.11: Result of Tyres 4

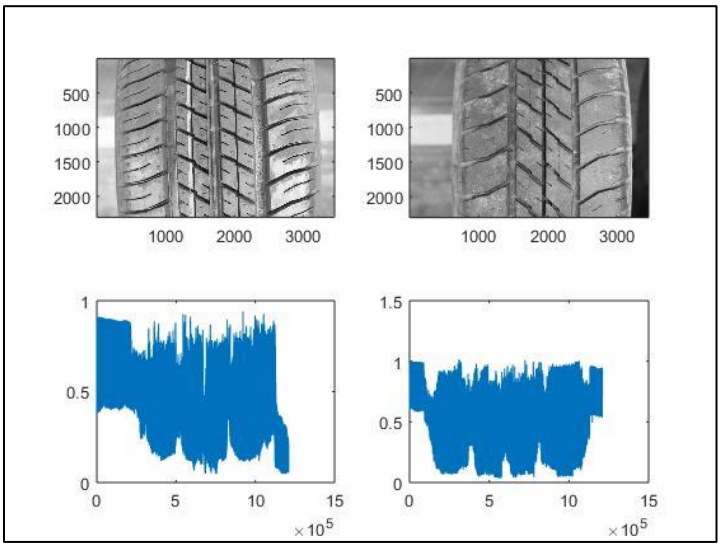


Figure 4.11: Result of Tyres 5

To summarize, in order to measure depth tread of the tyre by using Hausdorff distance, the algorithm first compute the distance between the image of the new and worn tyres. After that, the size function is used on the image of new tyres as well as worn tyres to return a row vector whose elements contain the length of the corresponding dimension of the new and worn image of the tyres. After the length of the corresponding dimension of the new and worn image of the tyres is calculated, for loop is used to loop through the all point looking for the absolute value while returns the length of dimension by comparing the image of new tyres and

worn tyres and plot the result of the algorithm. Next, the measurement of depth tread of tyre by using the absolute value calculated between the image of new and worn tyres is calculated by multiplier 10.

4.5 RESULT AND DISCUSSION

Table 4.2 shown the computed result for Algorithm 1, Algorithm 2 and Algorithm 3. The depth tread of tyres is measured in millimeter (mm) to achieve standardization with all measurement of tyres. The manual measurement of worn tyres is taken to in order to implement the accurate the algorithm to measure depth tread of the tyres. The measurement of depth tread of tyres is taken in four decimal places. The tyres are numbering based on file name as shown in Table 4.1 respectively.

Table 4.2: Depth Tread of Tyre for Each Algorithm in Millimeter

Tyres No	Manual measurement	Algorithms		
		Algorithm 1 Histogram	Algorithm 2 Edge Detection	Algorithm 3 Hausdorff Distance
1	4.65 mm	4.7210 mm	4.5689 mm	4.4717 mm
2	3.90 mm	4.0715 mm	4.2874 mm	6.2761 mm
3	4.75 mm	4.8830 mm	3.0436 mm	4.6928 mm
4	5.20 mm	5.3841 mm	4.0957 mm	5.5277 mm
5	3.40 mm	2.5944 mm	3.1757 mm	3.5937 mm
Accuracy (%)		87 %	55 %	85 %

Accuracy based on percentage difference: Algorithm 1 > Algorithm 3 > Algorithm 2

Table 4.3, 4.4 and 4.5 shown the computed result of depth tread of tyres in percentage as well as the condition of the tyres. The percentage and condition of the tyres is constructed based on Table 3.4.

Table 4.3: Percentage and Condition of Tyre for Algorithm 1

Tyres No	Depth Tread of Tyre	Percentage	Condition
1	4.7210 mm	47% worn	Tyre is in good condition.

2	4.0715 mm	47% worn	Tyre is in good condition.
3	4.8830 mm	47% worn	Tyre is in good condition.
4	5.3841 mm	31% worn	Tyres is in good condition.
5	2.5944 mm	78% worn	Inspect monthly.

Table 4.4: Percentage and Condition of Tyre for Algorithm 2

Tyres No	Depth Tread of Tyre	Percentage	Condition
1	4.5689 mm	47% worn	Tyre is in good condition
2	4.2874 mm	47% worn	Tyre is in good condition
3	3.0436 mm	62% worn	Tyre is in good condition
4	4.0957 mm	47% worn	Tyre is in good condition
5	3.1757 mm	63% worn	Tyre is in good condition

Table 4.5: Percentage and Condition of Tyre for Algorithm 3

Tyres No	Depth Tread of Tyre	Percentage	Condition
1	4.4717 mm	47% worn	Tyre is in good condition
2	6.2761 mm	15% worn	Tyre is in good condition
3	4.6928 mm	47% worn	Tyre is in good condition
4	5.5277 mm	31% worn	Tyre is in good condition
5	3.5937 mm	62% worn	Tyre is in good condition

By referring to Table 4.2, the highlighted measurement in the table is the measurement that is close to the manual measurement. Based on the accuracy of each algorithm, the accuracy of Algorithm 1 which is Image Comparison by Using Histogram is 87%, the accuracy of Algorithm 2 which is the Image Comparison by Using Edge Detection is the 55% and the accuracy of Algorithm 3 which is the Image Comparison Using Hausdorff Distance is 85%. From the accuracy, Algorithm 1 has the highest accuracy compared to Algorithm 2 and 3 and the lowest accuracy is Algorithm 2.

Based on the Algorithm 1, Image Comparison Using Histogram has the 87% accuracy on the measurement of tyres 1, 2, 3 and 4 while the measurement on tyres 5 is slightly different from

the manual measurement. The result of every measurement of tyres may be varied under several circumstances. From the result in Figure 4.5 of tyres 5, the Histogram showed a large spike at 173 intensity value of new tyres and 149 intensity of worn tyres. The image of the new tyre has the higher intensity compared to worn tyres because of the appearance of the line green and yellow on the new tyre. The external factor of color slightly affects the result of the measurement.

Based on algorithm 2, Image Comparison Using Edge Detection has the 55% accuracy on the measurement of tyres 1, 2 and 5 while the measurement of tyres 3 and 4 has a considerably large difference with the manual measurement. The large difference with manual measurement may cause by the Sobel Edge Detector. The Sobel operator is slower to compute and the operator less sensitive to noise. The operator also generally produces considerably higher output values for a similar edge.

Based on the Algorithm 3, Image Comparison Using Hausdorff Distance has the 85% accuracy on the measurement of tyres 1, 3, 4 and 5 while the measurement on tyres 2 has a considerably large difference with the manual measurement. The result of is difference may have caused by the position of the tyres in the image. The image of used tyres is not centered and titled a little bit compared to the image of new tyres. Therefore, when the image of new and used tyres is compared, difference result is computed because Hausdorff Distance measure the extent of every point appear on a model that agreed on some point of a related image set or vice versa.

The other factors that affect the measurement of depth tread of tyres by using digital image processing are caused by the uncontrolled environment during the image of the tyres is taken by using DSLR. The uncontrolled environment may be caused by insufficient light, the tyres are not centered, noise, and others.

4.6 IMPLEMENTATION OF THE GUIDE WITH MATLAB

The implementation of GUIDE or Graphical User Interface Development Environment is the last step involved in requirement analysis and design. Based on the result, the most accurate

result is Image Comparison Using Histogram compared to Edge Detection and Hausdorff Distance.

Therefore, Histogram algorithm is implemented in the GUIDE to measure the depth tread of the tyres. The GUIDE is implemented by using MATLAB and the algorithm of Histogram. Figure 4.11 shows the Image Comparison using GUIDE. The GUIDE begins by browsing the image of new tyres and worn tyres and by clicking the button of compare image, the result of depth tread of tyres remained, percentage depth tread of tyres and condition of tyres will be shown as in Figure 4.12.

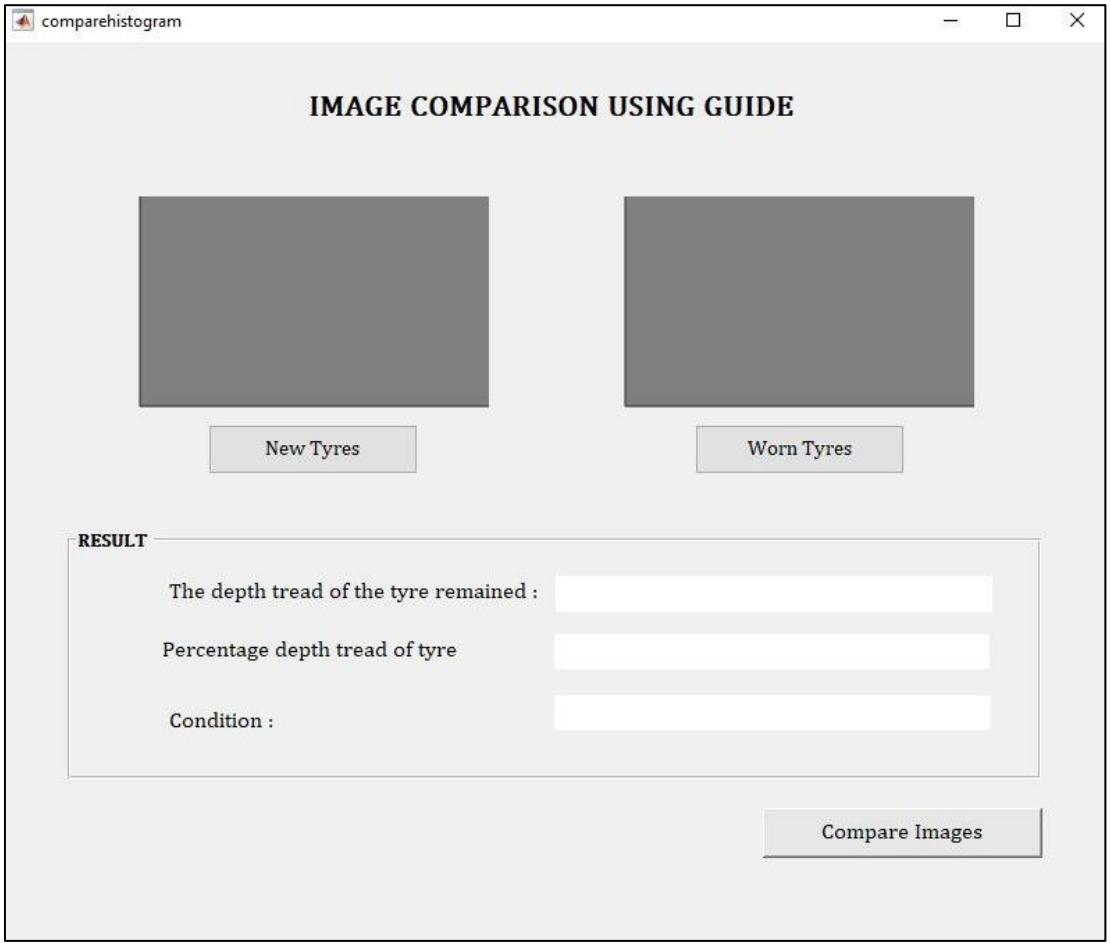


Figure 4.11: GUIDE for Image Comparison of Histogram

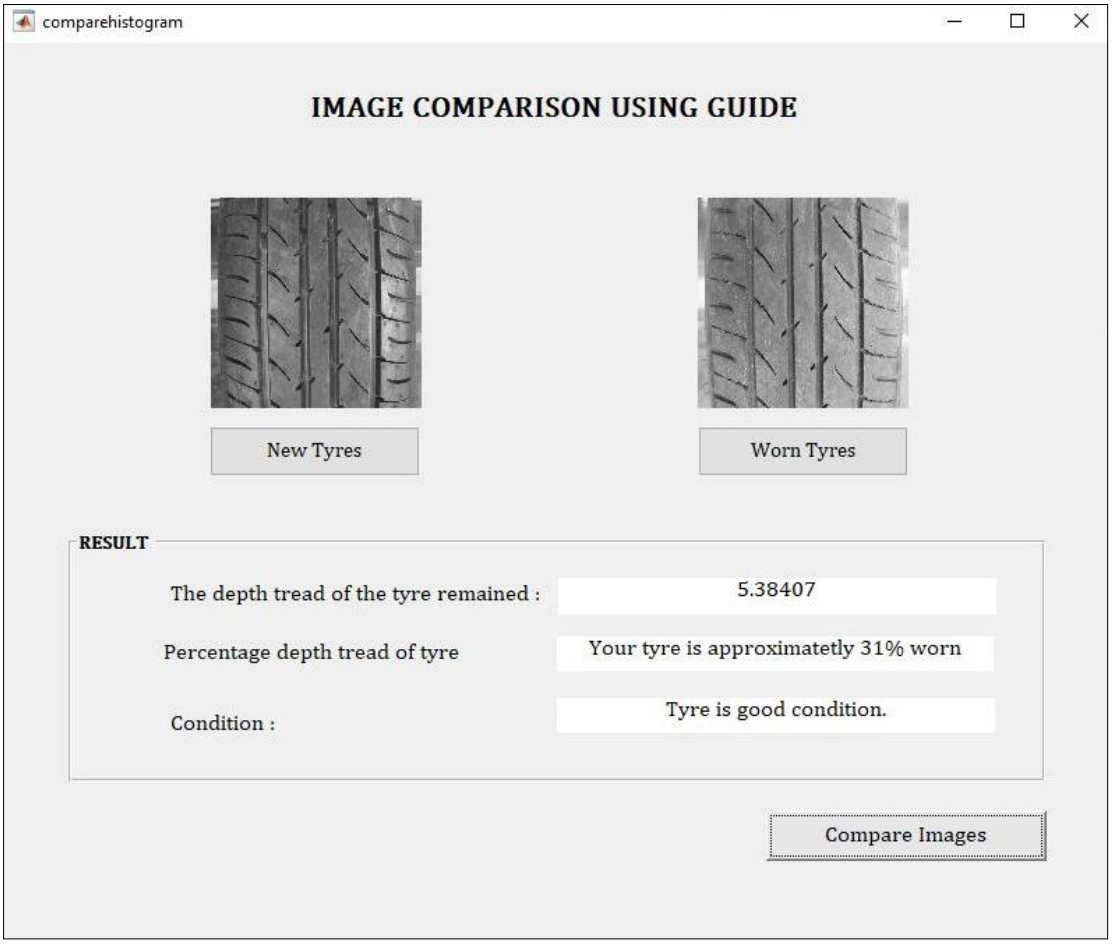


Figure 4.12: Result Displayed in the GUIDE

4.7 CONCLUSION

Finding the differences between two similar or identical images can be noticed by eye, but sometimes those differences are embedded and take a lot of time to be discovered, using some developed MATLAB code may make such operation rapid and accurate. Therefore, image comparison using Histogram has been chosen to measure depth tread of the tyres as it has the highest accuracy compared to the algorithms.

CHAPTER 5: CONCLUSION AND FUTURE WORK

5.1 INTRODUCTION

This chapter explained the overview of the whole achievement which has been obtained in conducting research in Tyres Tread Wear Indicator Using Digital Image Processing. In this final chapter, the algorithms are successfully implemented by using MATLAB R2015b. All the objective stated in Chapter 1 have been achieved successfully. The overall of this research-based project has been summarized in this chapter such as objective achievement, innovation and benefits, recommendation for future work, limitation in algorithm and hardware and also the conclusion of the overall chapter.

Moreover, at the end of this research-based project, this research-based project of Tyre Tread Wear Indicator Using Digital Image Processing can turn into commercial value and can be marketed by replacing the conventional technique of measuring depth tread of tyres by using depth gauge. Besides that, this project can also generate income for the developer if they turn this project into a mobile application or integrated this application as a new feature in the car which will assist users to get the tyres measurement in a short time.

Furthermore, in term of social value, this research-based project can improve safety awareness to the society as the implementation of the algorithm can help the user to alert and retain the depth tread of the tyres in the range of standard safety measure. The increment of the number of the accident is such a worrisome as it increased day by day and involved all ages of Malaysian's nations. As a computer science student, it would be my responsibility to contribute my knowledge in this field to decrease the number of the accident by implementing the suitable algorithm which can determine the depth tread of the tyres. The implementation of this algorithm can perform more accurate and efficient while providing a very safe's vehicle's condition on tyres. This implemented algorithm is more protective and introduces safer tyres conditions that can help in reducing the numbers of accidents happened every day caused by worn tyres.

5.2 OBJECTIVE ACHIEVEMENT

The objective achievement ensures that the implemented algorithms able to determine the depth tread of the tyres by comparing the image of depth tread of the new tyres and the worn tyres by using computerizing inspection. The Tyre Tread Wear Indicator Using Digital Image Processing can accurately measure the percentage of depth tread of the tyres and alert the user to retain a tread depth of the tyres in the range of standard safety measure. Several objectives have been stated in Chapter 1 such as:

- To propose an algorithm to measure depth tread of tyres by comparing three set of algorithms in digital image processing.
- To differentiate the depth tread between the used tyre and the new tyre by using digital image processing.
- To find out the best technique or algorithm to analyses the depth tread of the tyre by using digital image processing.
- To give an accurate measurement of depth tread of the tyre by using digital image processing.

Based in the above objective stated, the implemented algorithms in Tyre Tread Wear Indicator Using Digital Image Processing has achieved the entire objectives which provide an effective solution to determine the tread of tyres effectively by using computerizing inspection from time to time with an accurate measurement.

5.3 INNOVATION AND BENEFITS

Tyre Tread Wear Indicator Using Digital Image Processing can be considered as an innovation project in determining depth tread of the tyres by using computerizing inspection with providing an accurate measurement of the depth tread of the tyres. However, most of the technique in the market usually used conventional technique of tyres tread depth such as 20 cent coins test and depth gauge. This type of inspection is not very noticeable, inefficient and unsafe to the user which has higher chance or probability of the misjudgment and human error in getting the accurate measurement of the depth tread of the tyres as mentioned in Chapter 1.

Several innovation and benefits offered by the implemented algorithm in Tyre Tread Wear Indicator Using Digital Image Processing such as:

- The implemented algorithm able to determine the depth tread of the tyres and eventually can reduce the number of accidents caused by worn tyres.
- Able to give a percentage of the depth tread of the tyres accurately by using computerized inspection.
- The proposed technique provides more cheap and affordable computerized inspection.
- The implemented algorithm allows the images of tyres taken statically by using the camera without need the user to inspect manually.
- Provides safety measurement more accurate and effective as it can alert the user to retain a tread depth of the tyres in the range of standard safety measurement.
- The proposed technique is simple and uses simple algorithms to perform the operation of image processing towards the tyres in a short time.

5.4 LIMITATION

Every study will be bound by certain limitations that cannot be controlled and any limitations might influence the result of the study. In this research-based project, there are several limitations of Tyre Tread Wear Indicator Using Digital Image Processing such as limitation in implemented algorithm and hardware.

5.4.1 ALGORITHM LIMITATION

Although the algorithm can determine the depth tread of the tyres by using computerized inspection and can give an accurate result, there is still some limitation where there are rooms for incessant and continuous improvement in the future. This research-based project is initiated and due to the limitation of time, there is some part of that algorithm failed to investigate. In this project, the implemented algorithm can determine the depth tread of the tyres by comparing the new and worn images of the tyres and use the selected algorithm to find the percentage difference between these two images. However, the algorithm failed to measure the gradation happened between the new and worn images of the tyres.

The solution to this problem is to use the same tyres of one car and capture the image of the tyres from one time and to time and re-test the image by using the implemented algorithm or technique to stimulate how the tyres degrade over time.

5.4.2 HARDWARE LIMITATION

Based on Chapter 3, the hardware used to capture images of tyres is by using Digital Single-Lens Reflex (DSLR) model Canon EOS 50D. Even though DSLR has the advantage in lens selection, better optically viewfinder that works better in low light and can help to reduce the amount of unwanted information such as noise, DSLR is not an attachable device that can be attached permanently near to tyres.

Therefore, the solution to this problem is to use existing wireless Charged Coupled Device (CCD) to replace the DSLR. CCD are sensors used in digital cameras to record still and moving camera. The CCD captures light and converts it to digital data that is recorded by the camera. The wireless CCD can automatically transmit the data to the computer and can be inspected computerizing after receiving the data. The CCD is attachable, and way faster compared to DSLR because if using DSLR we must transmit data by using a card reader. By using CCD, we can capture the same model of the tyres even though, the cars are moving.

5.5 RECOMMENDATION FOR FUTURE WORKS

Firstly, this research has presented correspond to the final year project for the degree on a research study, by the fact we only focusing in implementing the new algorithm to measure the depth tread of the new tyres instead of improving and combining the existing technique with the implemented algorithm. Clearly, this research is focusing on the measuring the depth tread of the tyres by using digital image processing. There are two recommendations for future work under this research-based project which are:

- Implements of a mobile application that can directly inform the user the remaining depth tread of the tyres when capturing the image of the tyres by using smartphone.
- Integrated the Tyre Tread Wear Indicator Using Digital Image Processing as a new feature in cars.

Even though the implemented algorithms can measure the depth tread of the tyres by using comparing techniques between the new and worn image of the tyres, the gradation technique is recommended as it can be more accurate in measuring the depth tread of the tyres if the gradation technique is implemented in the current algorithms.

In the same time, implementation of a new application as shown in Figure 5.1 to measure the depth tread of the tyres by capturing the tyres by using smartphone is also recommended. This type of application can help the user to alert the user to retain a tread depth of the tyres in the range of standard safety measure. It can also assist the user to know the current measurement of the depth tread of tyres faster by capturing the images of the tyres and the application will process computerizing and display the output or the result in the short times.

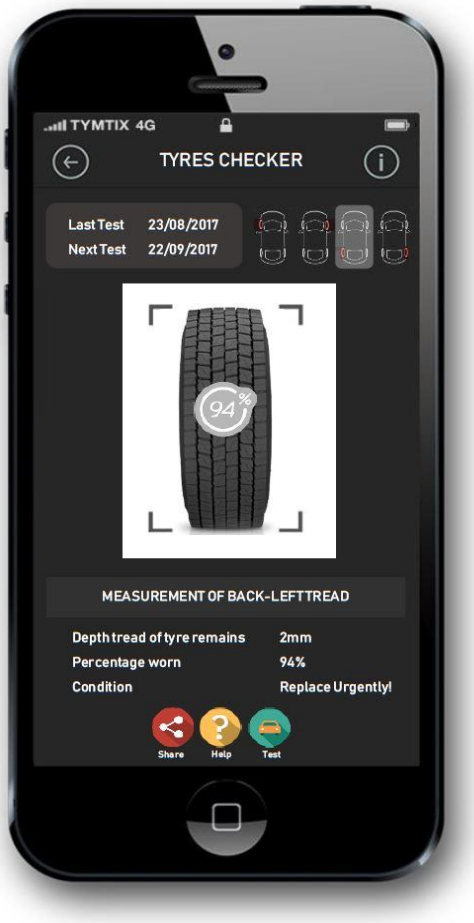


Figure 5.1: Application of Tyres Checker

Last but not least, integrating the Tyre Tread Wear Indicator Using Digital Image Processing as a new feature in cars. The wireless CCD camera can be placed above the four tyres and the user can monitor all the four tyres condition on the car's dashboard at one time as shown in Figure 5.2. By integrating this feature, it can alert the user to retain a tread depth of the tyres in the range of standard safety measure and this is a good way to decrease the number of accident in Malaysia.

Figure 5.2: Four Tyres Condition on Dashboard

5.5 CONCLUSION

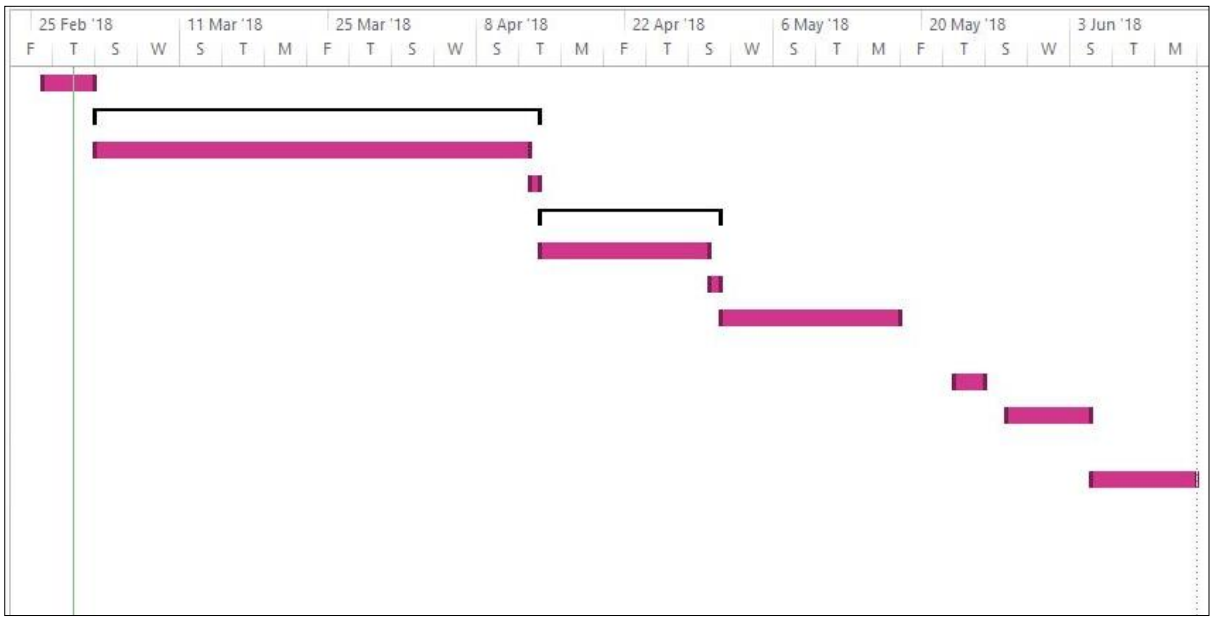
In this final year project, the main purpose of this research-based was to determine the algorithm to measure depth tread of the tyres by using digital image processing or computerized inspection. The algorithms used can help to decrease the statistical number of accident in Malaysia which caused by the worn tyres.

Lastly, the algorithms in this research-based project were successfully developed based on objective stated in this project. The improvement can still be done to develop an effective technology measurement of the depth tread of the tyres that can really fulfill user needs and requirement.

APPENDIX A: GANTT CHART

Gantt chart for Final Year Project 1

Gantt chart for Final Year Project 2



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